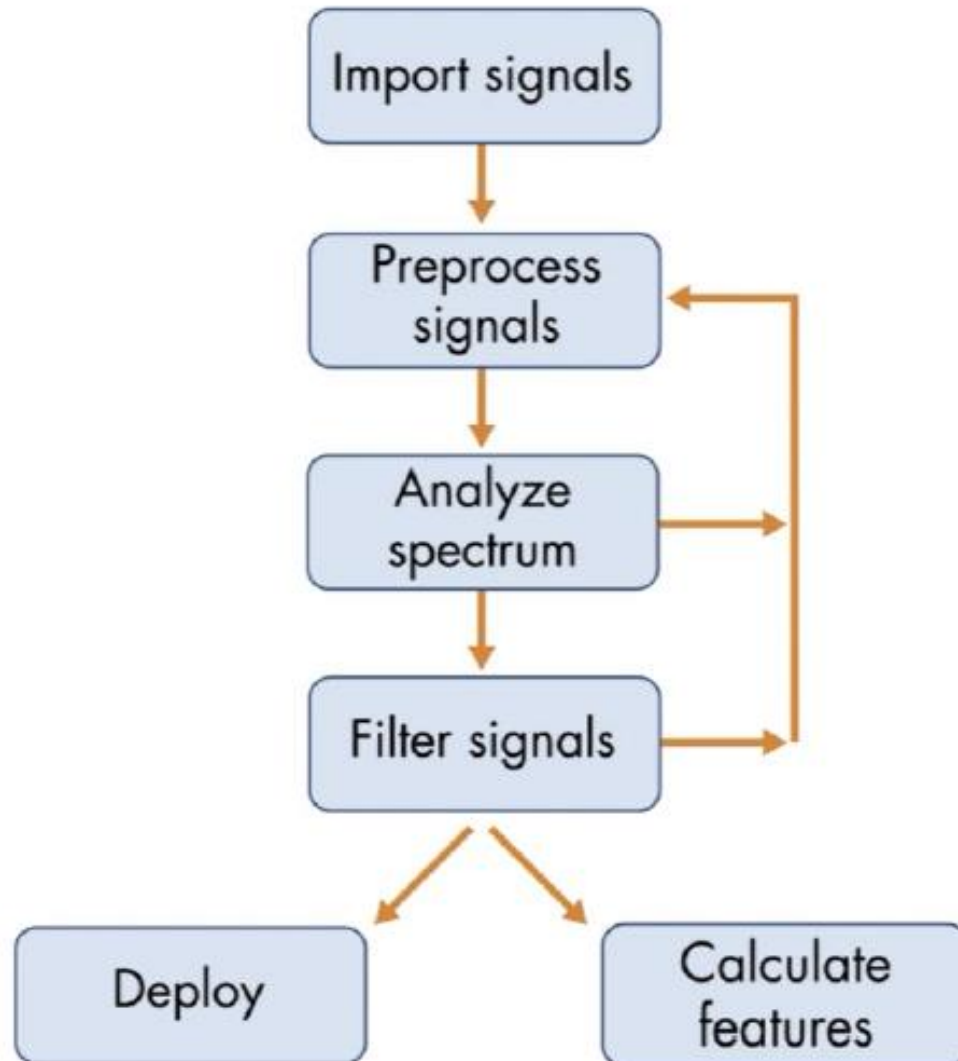


Signal Processing Using MATLAB

18th Feb 2024

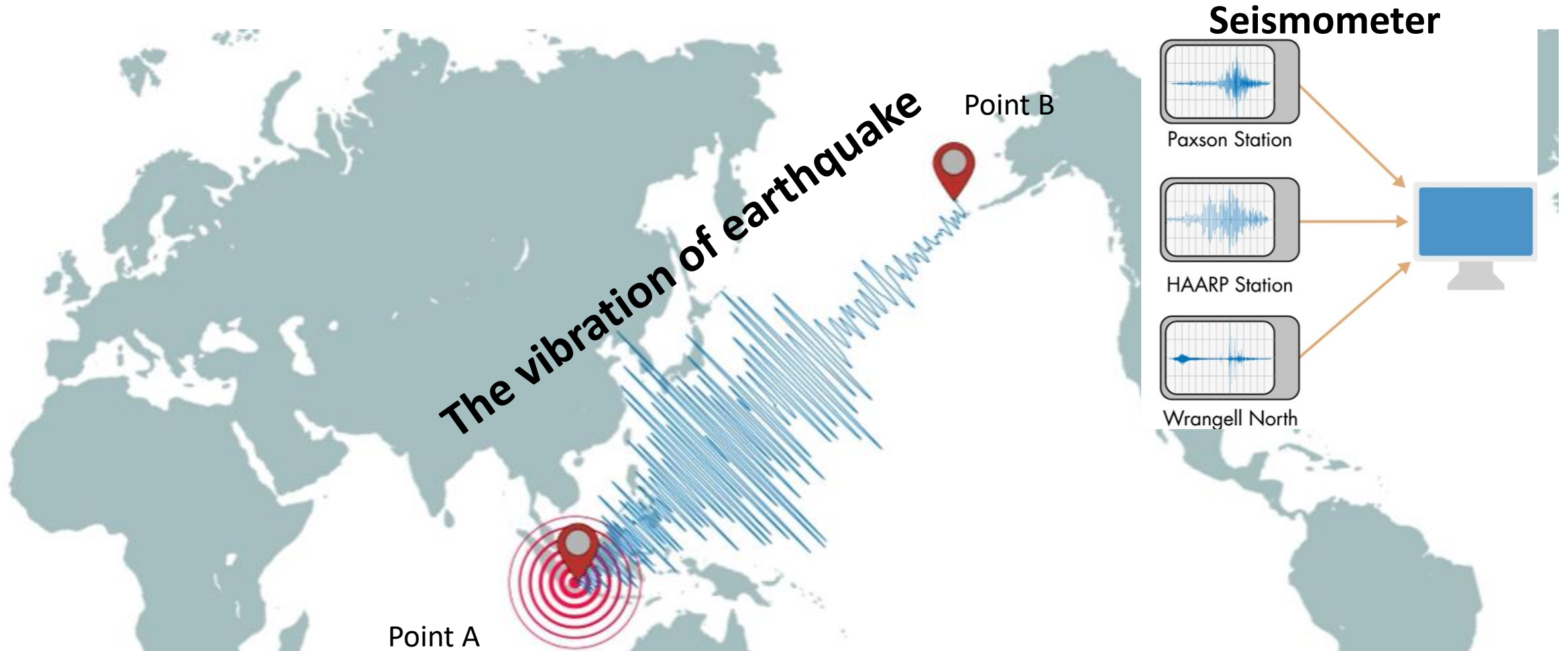
Dr. Saif Ali Jasim Al Grait

Overview: Signal Processing Workflow



1. Import the signals into MATLAB
2. Preprocess the signals in the time domain to prepare them for analysis
3. Look at the power spectrum in the frequency domain
4. Filter the signals to keep only relevant frequency content
5. Revisit earlier steps in the process to refine your algorithm
6. Extract information from the signals or deploy your algorithm to hardware

Task 1: Perform signal processing for earthquake



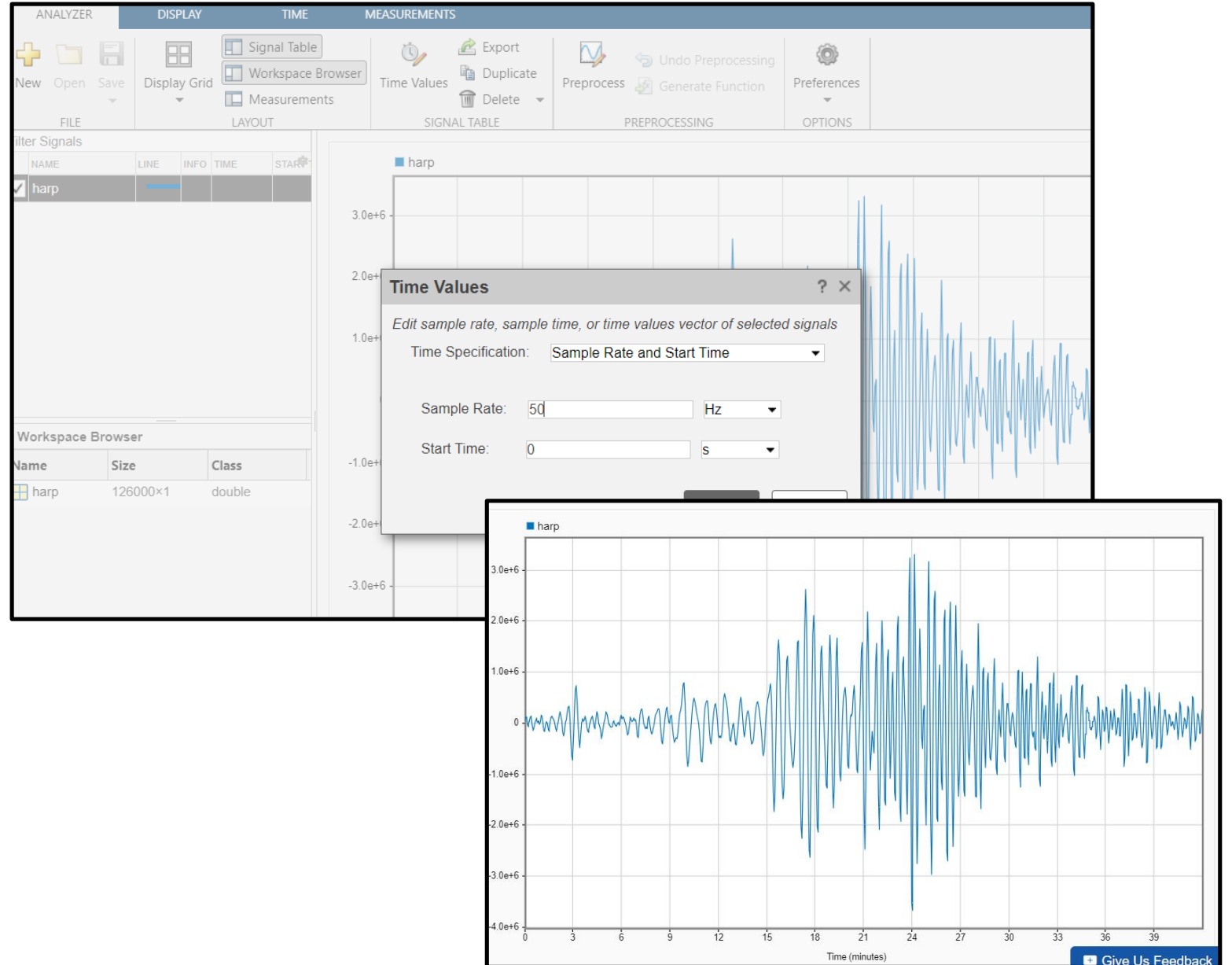
Set time information in spectrum analyzer

To set the time information, select  **Time Values** in the **Analyzer** tab. You can specify the sample rate (f_s), the time between each sample ($\frac{1}{f_s}$), or a vector of time values.

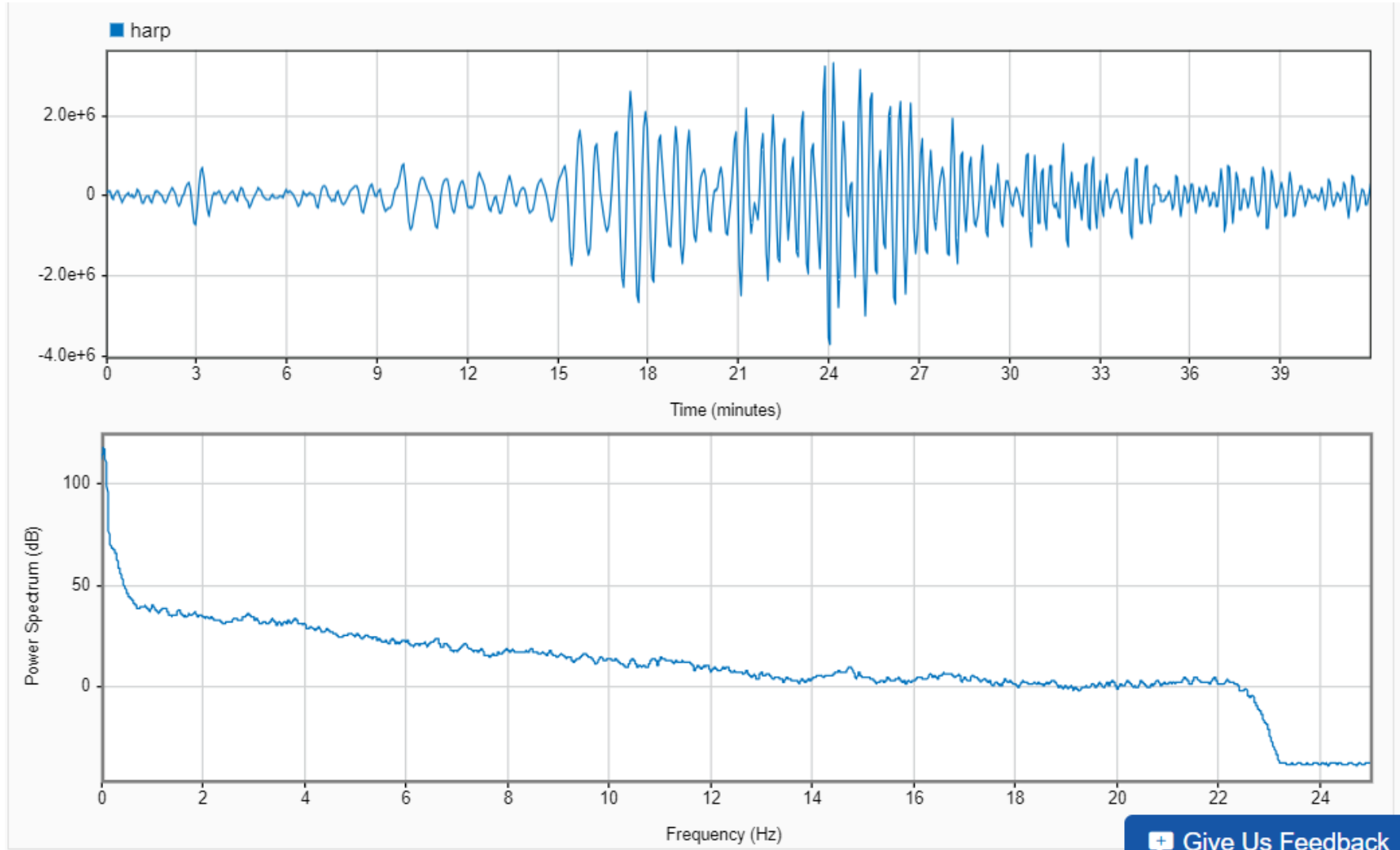
TASK

Set sample rate to Hz.

In the Time Specification drop-down, select Sample Rate and Start Time. Keep the default Start Time of 0 seconds.



To view the frequency content of a signal, you can display its power spectrum by clicking Spectrum on the Display tab



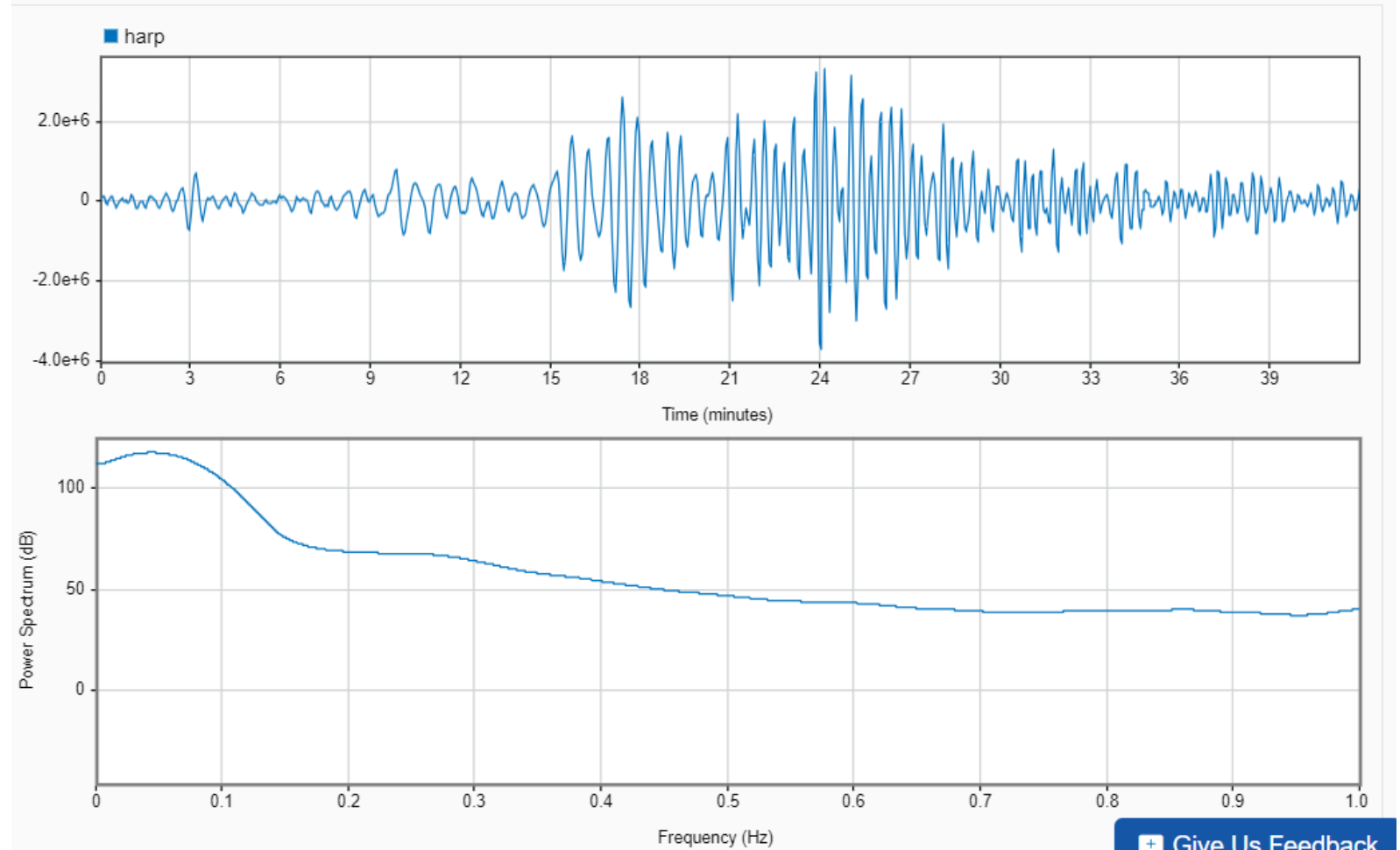
To view the frequency content of a signal, you can customize the visualization

While you display the power spectrum, a **Spectrum** tab contains options to customize the visualization.

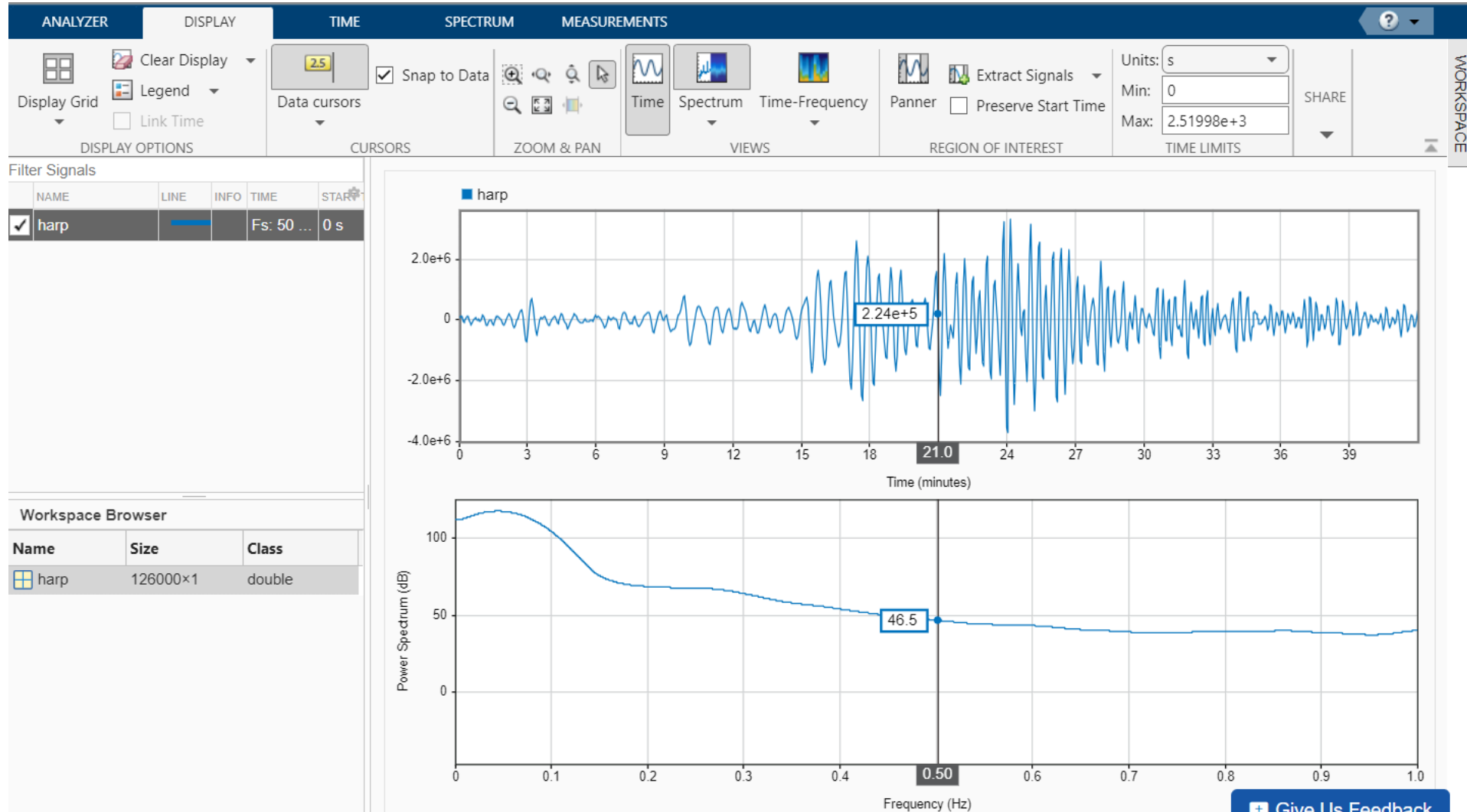
The frequency range of these seismic surface waves is small, so to focus on the meaningful seismic activity, you can set the frequency limits of the power spectrum.

TASK

Set the frequency limits from Hz to Hz.



To view the frequency content of a signal, you can add data cursor

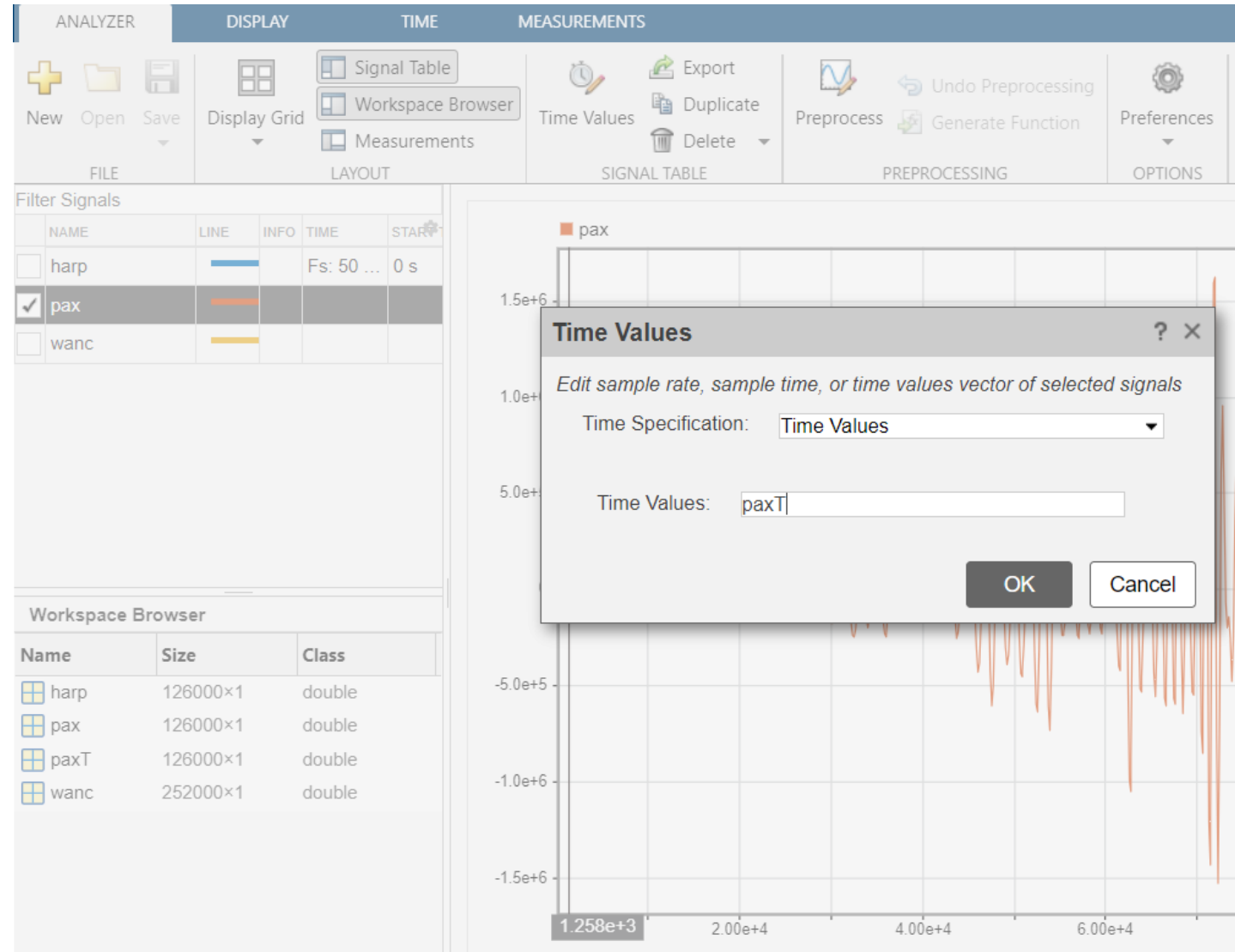


Compare signals

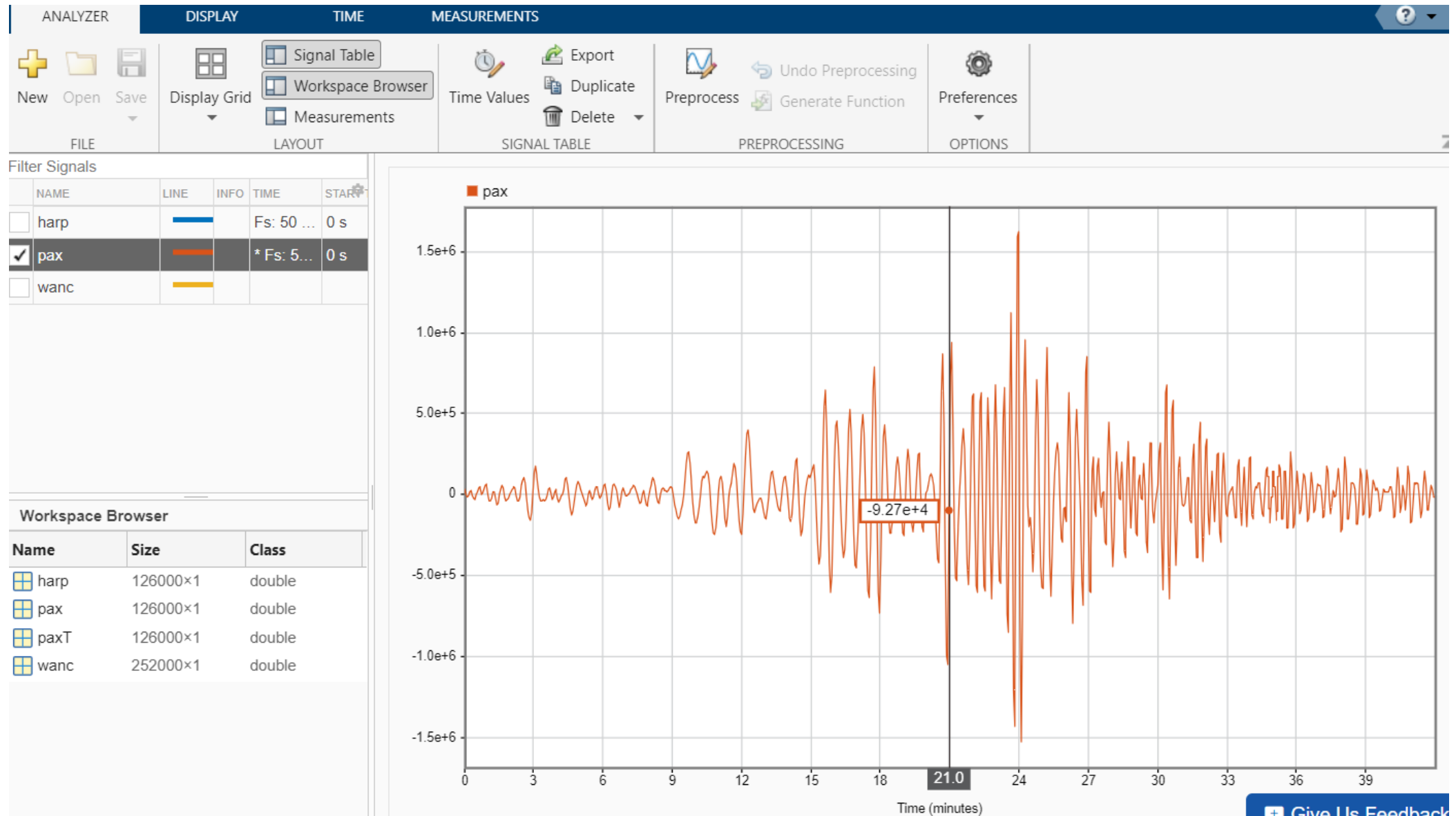
The three seismic stations HARP, PAX, and WANC collected the seismic signals using different hardware. As you've seen, the data from the HARP station was sampled at 50 Hz.

The data from the PAX station was also sampled at 50 Hz, but the hardware was imperfect, so the spacing between each sample wasn't exactly 0.02 seconds. Since the time information of the signal isn't evenly spaced, the time information at every sample needs to be specified.

You can specify a vector instead of a sample rate using the **Time Values** time specification.



Compare signals



Compare signals: set the sample rate of WANC station

The signal from the WANC station is uniformly spaced, but it was sampled at a faster sample rate of 100 Hz.

TASK

Set the sample rate of the `wanc` signal to 100 Hz and display the signal.

The screenshot shows a software interface with a 'Time Values' dialog box open. The dialog box has a title bar with a question mark and a close button. Below the title bar, it says 'Edit sample rate, sample time, or time values vector of selected signals'. The 'Time Specification' dropdown is set to 'Sample Rate and Start Time'. The 'Sample Rate' is set to 100 Hz, and the 'Start Time' is set to 0 s. There are 'OK' and 'Cancel' buttons at the bottom right of the dialog box.

The background interface shows a 'Signal Table' with the following data:

NAME	LINE	INFO	TIME	START
harp			Fs: 50 ...	0 s
pax			* Fs: 5...	0 s
wanc			Fs: 10...	0 s

Below the 'Signal Table' is a 'Workspace Browser' table:

Name	Size	Class
harp	126000×1	double
pax	126000×1	double
paxT	126000×1	double
wanc	252000×1	double

The main display area shows a plot of the 'wanc' signal, which is a yellow waveform on a grid. The y-axis ranges from -9.0e+6 to 9.0e+6, and the x-axis ranges from 0 to 10.

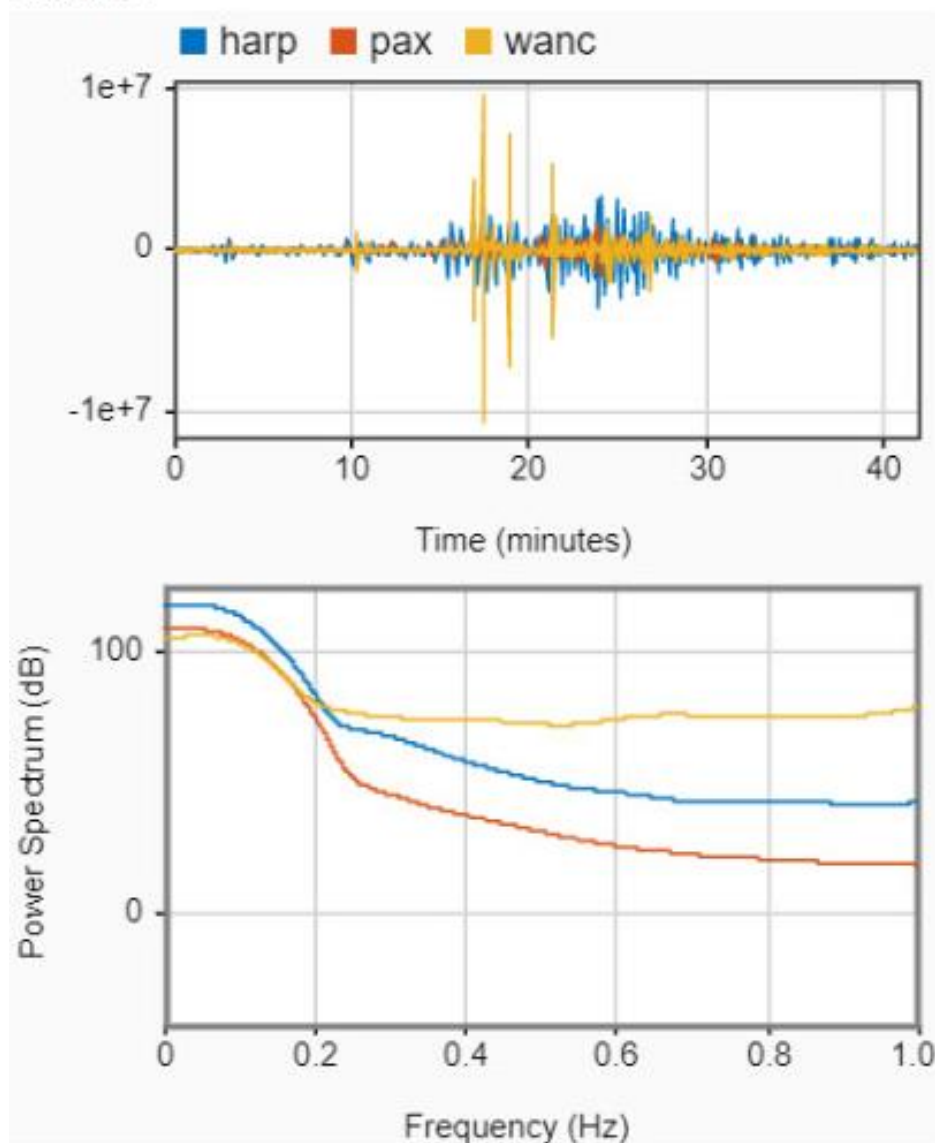
Compare signals, it is easily to compare them in frequency domin

The different signal amplitudes make it difficult to compare the signals in the time domain, but you can still compare them in the frequency domain.

TASK

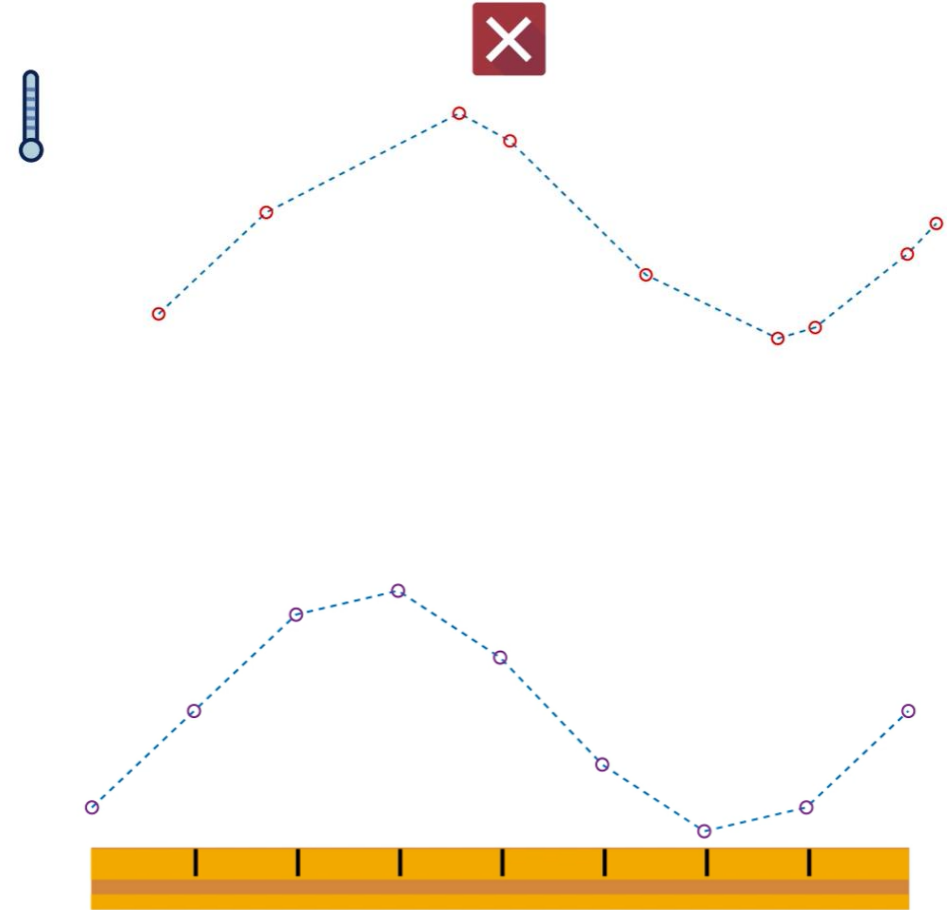
Display all three signals, then display the power spectra. Set the frequency limits from 0 to 1 Hz.

ANALYZER		DISPLAY	TIME	SPECTRUM	MEASUREMENT
Units:	Hz				
Min:	0	Min:	-4.295646141e+1	Frequency Scale:	Linear
Max:	1	Max:	1.258956540e+2	☒ Spectrum in dB	
FREQUENCY LIMITS		Y-AXES LIMITS		SCALE	



Imperfect sensor usually generates imperfect data

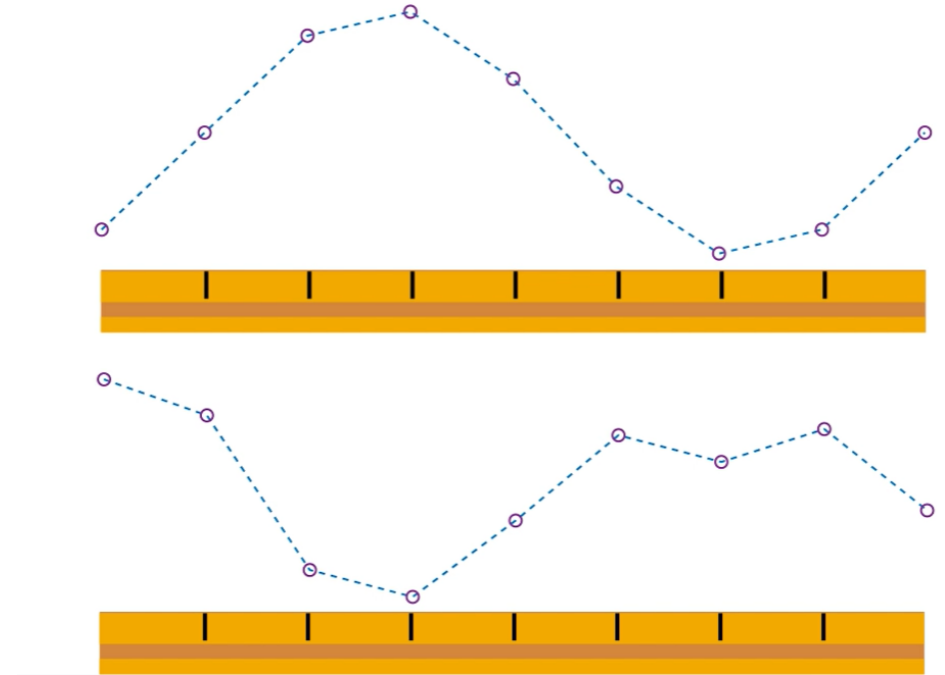
- Perfect can be obtain when we generates the signals
- Basically every real world data are imperfect, thus
- It needs preprocessing before we can reuse it
- Most signal processing technique assumes you have uniformly sampled data



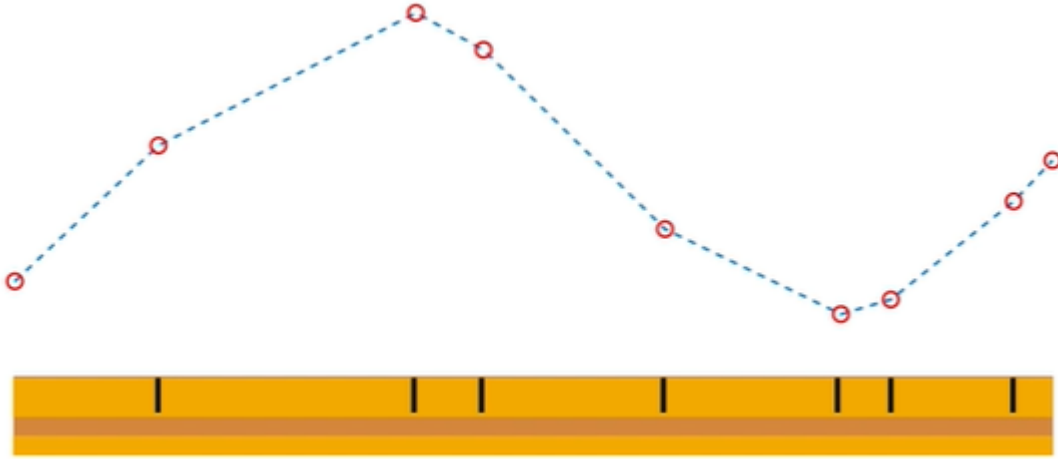
Signals standardization

- When we have multiple signal, the timestamps must match up

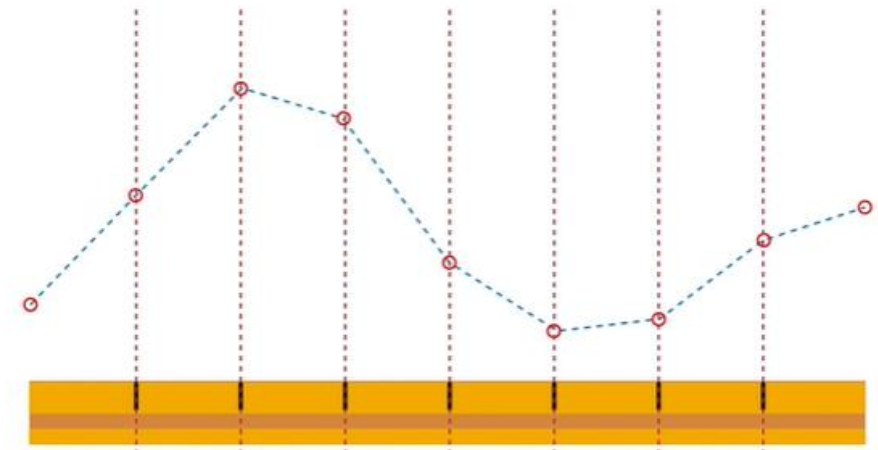
So how we standardize our signals?



Non-uniformly sampled signals don't have equal spacing between each point

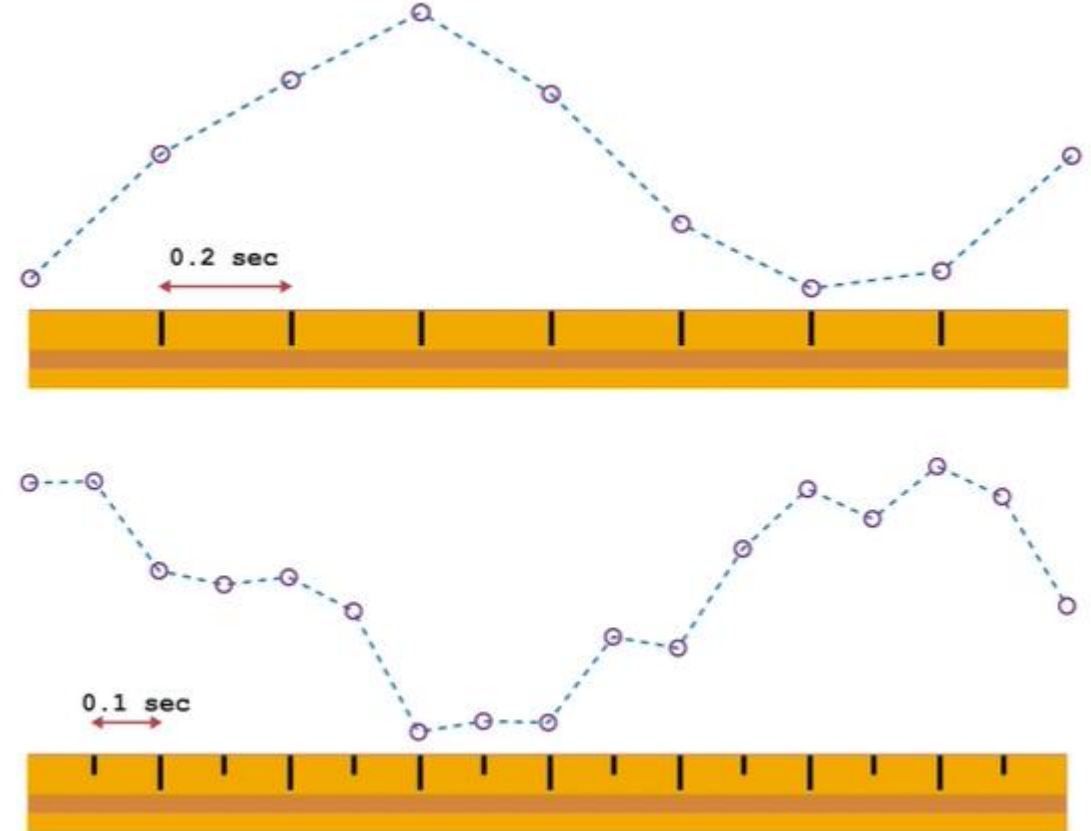


- Signals can be interpolated to new equally spaced time steps using **resampling**
- Once all signals are uniform, it's time to line up their time stamps



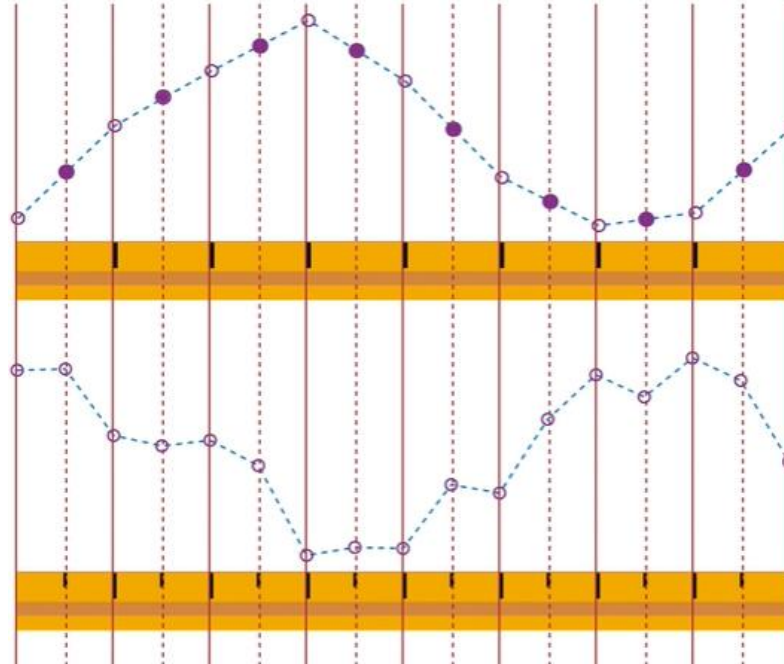
Non-uniformly sampled signals don't have equal spacing between each point

- When you have multiple signals, they might be samples at faster or slower rates
- So, the sample rate can be changed of one of the signals to match another

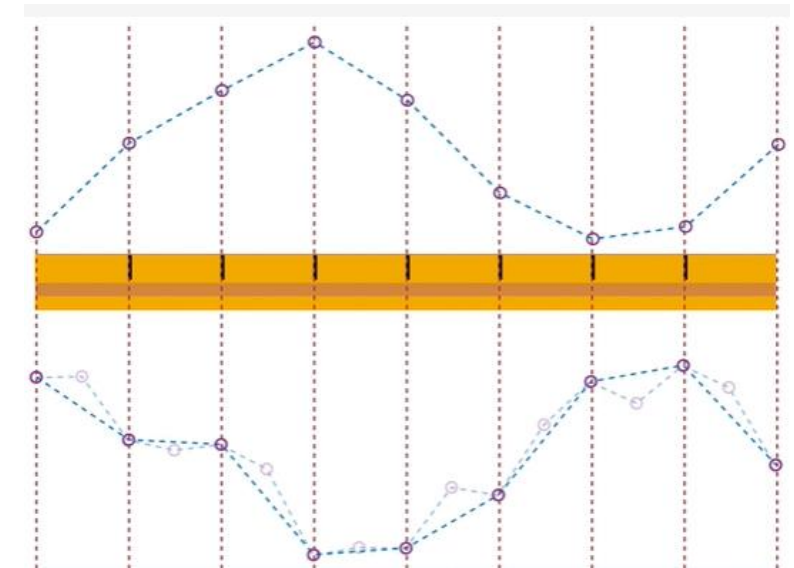


Generates new information by adding new samples to a signal

- One way is by up-sample a signal to a faster sample rate over interpolating
- “Interpolation is a process of determining the unknown values that lie in between the known data points”



- Or you can down-sample a signal to a slower sample rate

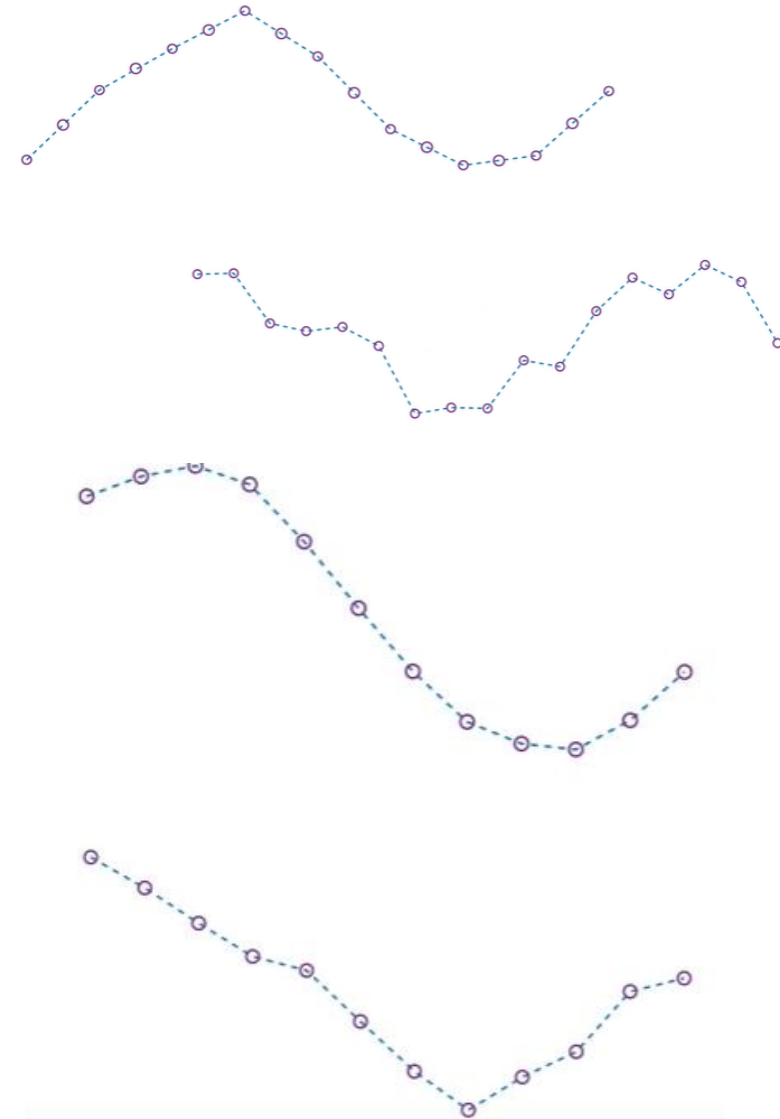


The highest frequency information you have is limited by the slowest sample rate by down-sampling you throwing away the high frequency information

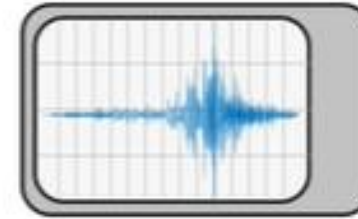
However, by up-sampling , you don't gain high-frequency information because you just interpolating

Once the time base is cleaned up, the signal itself might also need some preprocessing

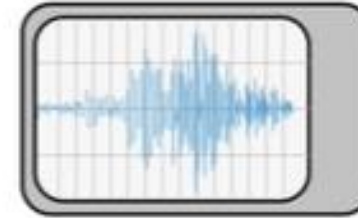
There are techniques to remove distracting patterns like irrelevant trends and noise



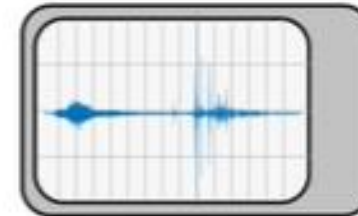
So, we need to align the earthquakes signals from different stations



Paxson Station



HAARP Station

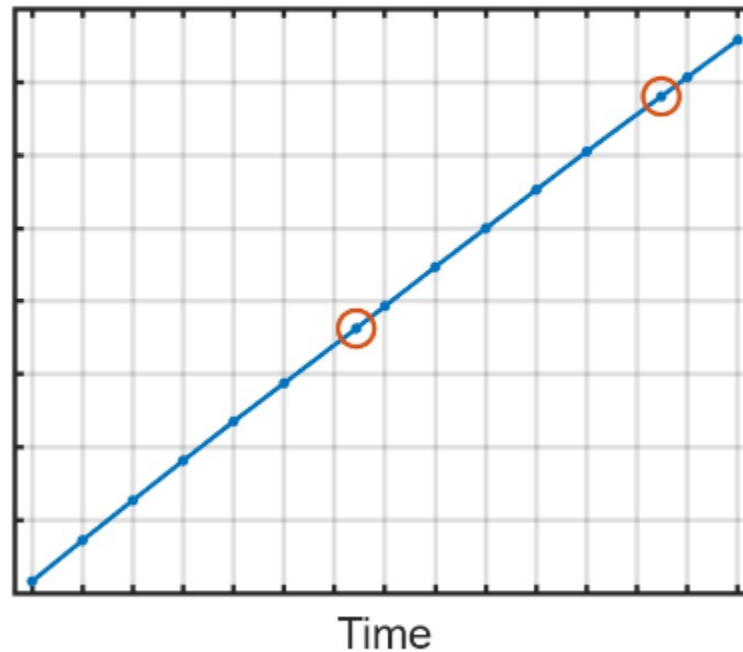


Wrangell North
Crater Station

Spectrum analysis techniques requires signals to be uniformly sampled

Nonuniform Sampling

The signal from the PAX seismic station is not uniformly sampled. Uniformly sampled signals have equal spacing between every point, but some of the samples in the `pax` signal were recorded with a small delay.

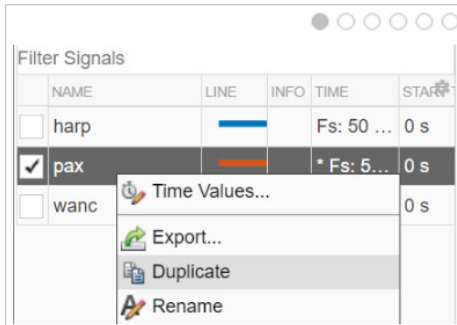


Most spectral analysis techniques require signals to be uniformly sampled, so Signal Analyzer interpolates the signal to a uniform grid when you view the power spectrum. To establish a consistent time base for all three signals, you can resample the `pax` signal to 50 Hz.

Before you modify a signal, it is best practice to duplicate the signal so that you can compare the signal before and after processing. To process a signal, follow these steps:

Background

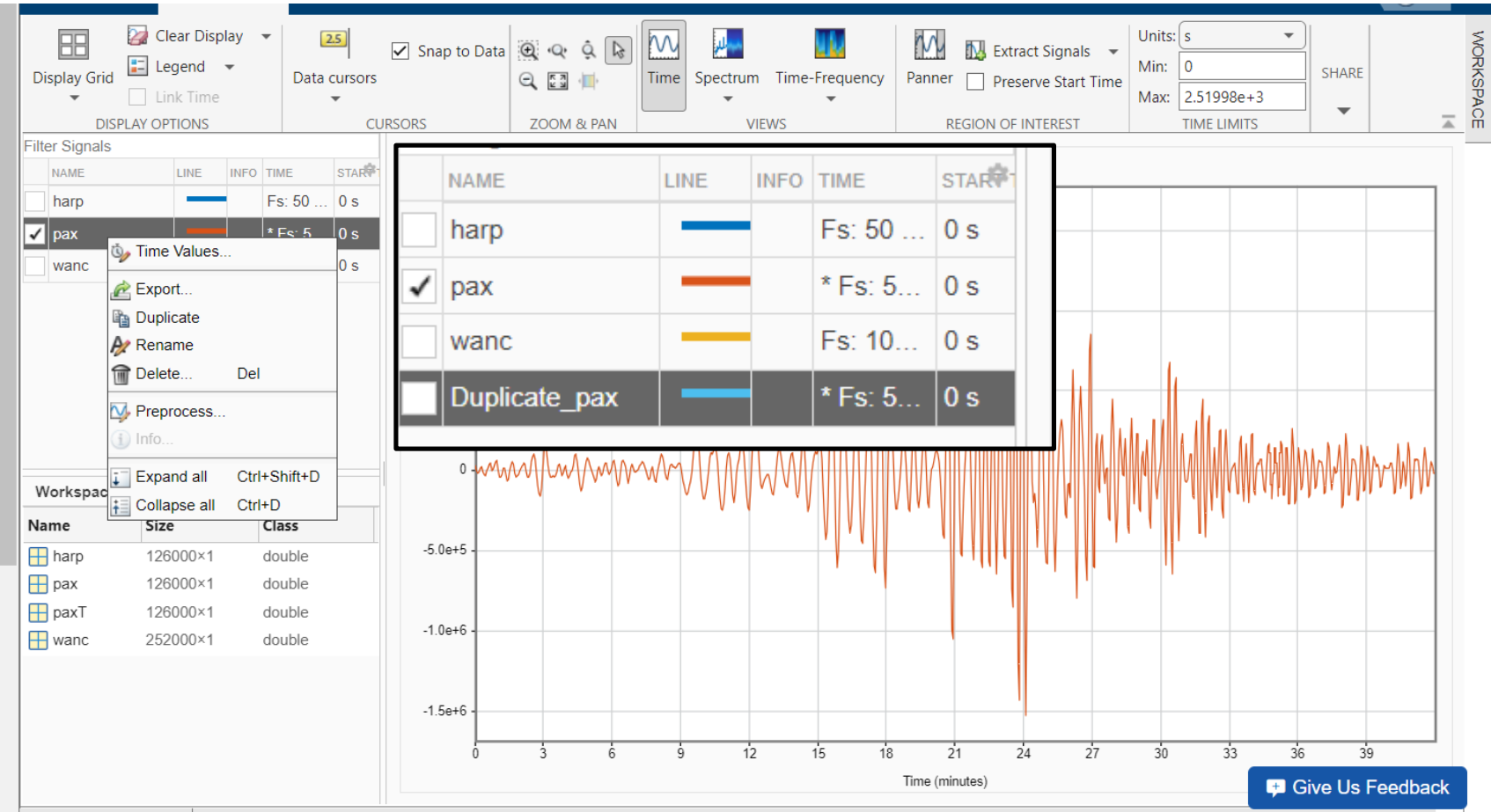
Before you modify a signal, it is best practice to duplicate the signal so that you can compare the signal before and after processing. To process a signal, follow these steps:



Duplicate or rename the signal by right-clicking it in the Signal table and selecting **Duplicate** or **Rename**.

TASK

Duplicate the `pax` signal and name the new signal

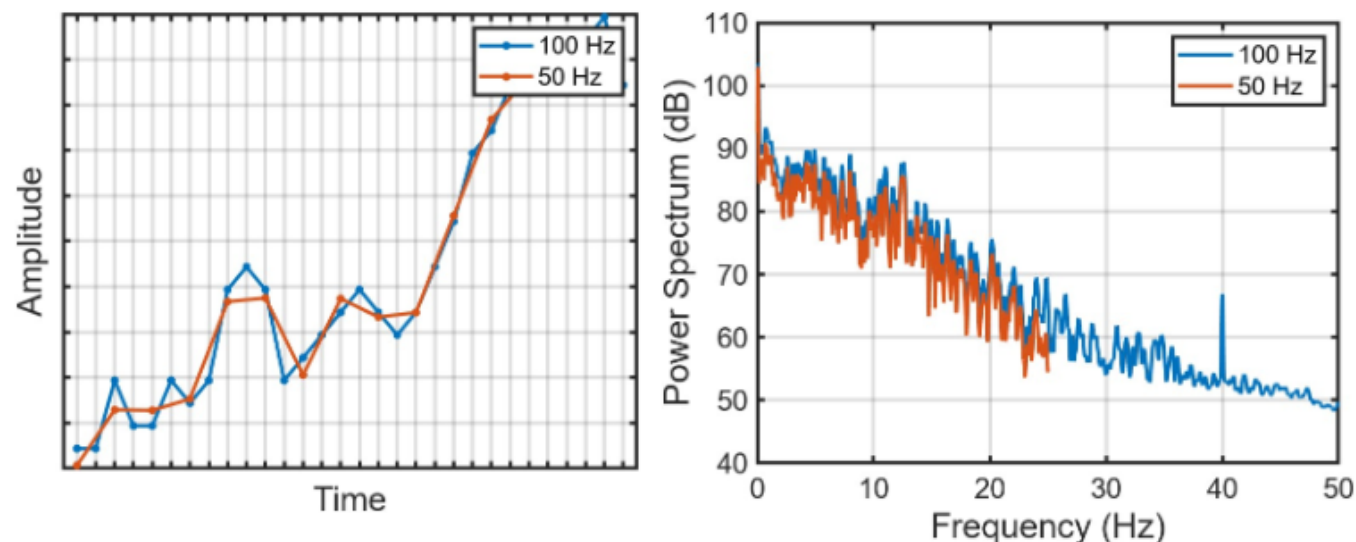


Different Sample Rates

When you use the proper tools to interpolate a nonuniform sample to a uniform grid, the power spectrum of your signal shouldn't noticeably change. But if you change the sample rate, there is always a change in the power spectrum.

The sample rate determines the highest frequency that can be correctly represented in a sampled signal. So before you change the sample rate, you should look at the power spectrum to see where the frequencies are located.

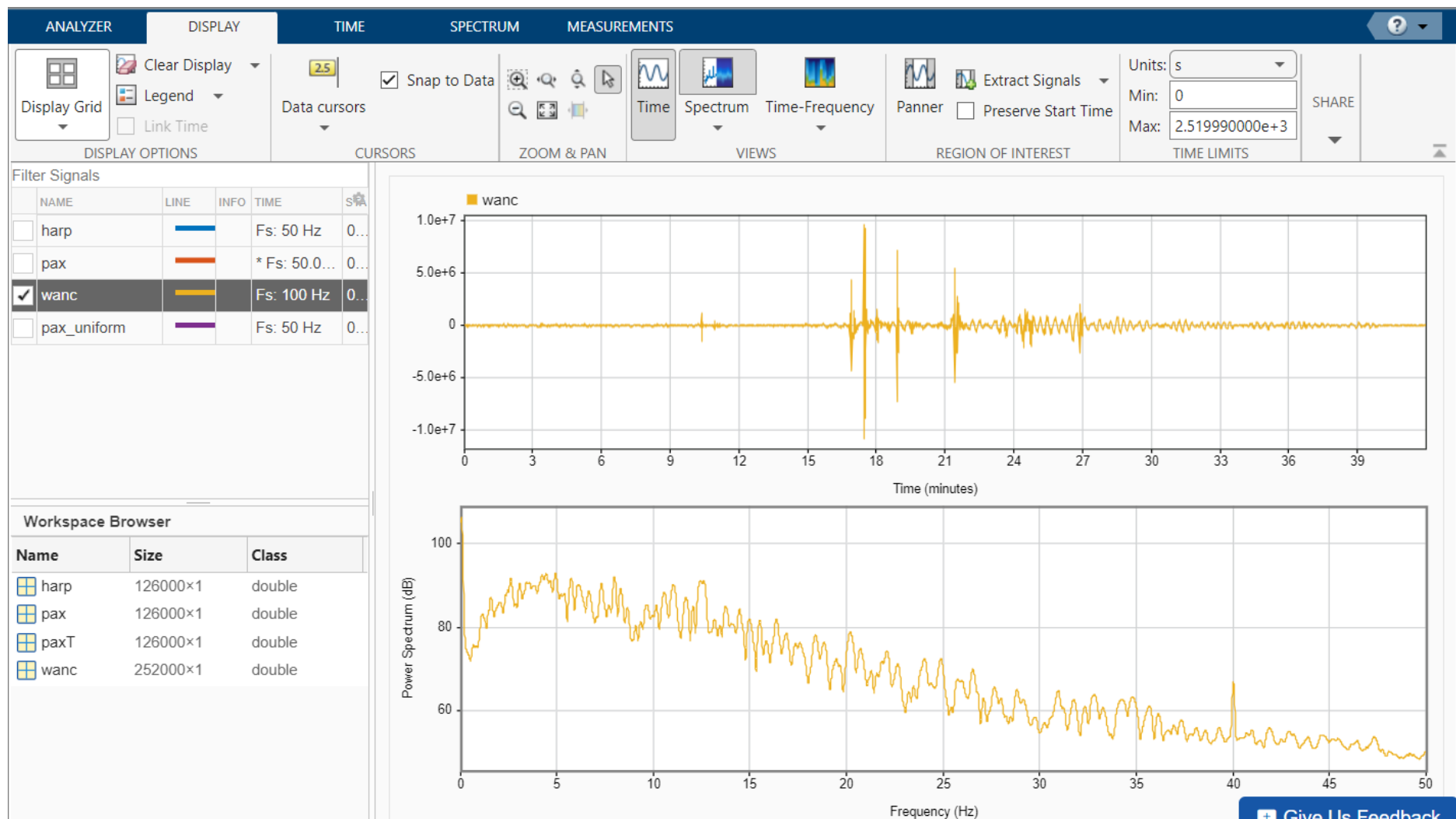
$$f_{max} = \frac{f_s}{2}$$



In the example above, the sample rate decreased from 100 Hz to 50 Hz. In the 100 Hz signal, before resampling, you should pay attention to the frequency content from 50 Hz to 100 Hz. Most of the power is between 0 and 50 Hz. In the 50 Hz signal, after resampling, the frequency content between 0 and 50 Hz isn't significantly different from the original signal.

Different Sample Rates

- The signal recorded at the Wrangell station has a faster sample rate of 100 Hz
- Look at the full frequency range of the wanc signal's power spectrum.



Different Sample Rates

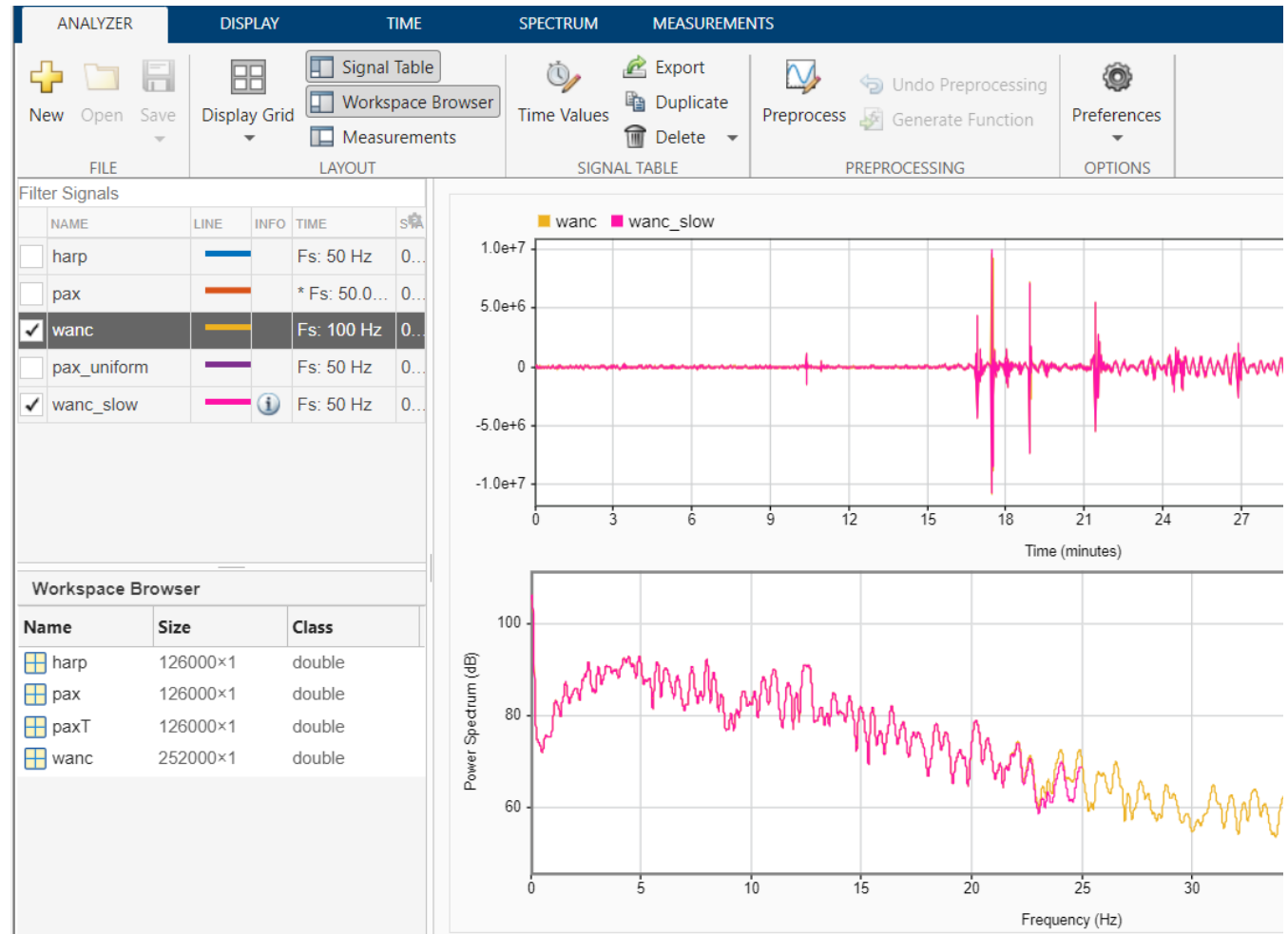
When you resample to a slower sample rate, frequency content above $\frac{f_s}{2}$, where f_s is the sample rate after resampling, can distort the power spectrum.

In the WANC signal, there isn't significant power above 25 Hz, so you can try resampling the WANC signal to 50 Hz so that it matches the sample rate of the HARP and PAX signals.

TASK

Duplicate `wanc` and rename the copy `wanc_slow`. Then resample `wanc_slow` by a **Sample Factor** of `0.5`.

Display the original and resampled signals to compare the effects of resampling.



Resample: Resample uniform or nonuniform data to new fixed rate

Syntax

```
y = resample(x,p,q)
y = resample(x,p,q,n)
y = resample(x,p,q,n,beta)
y = resample(x,p,q,b)
```

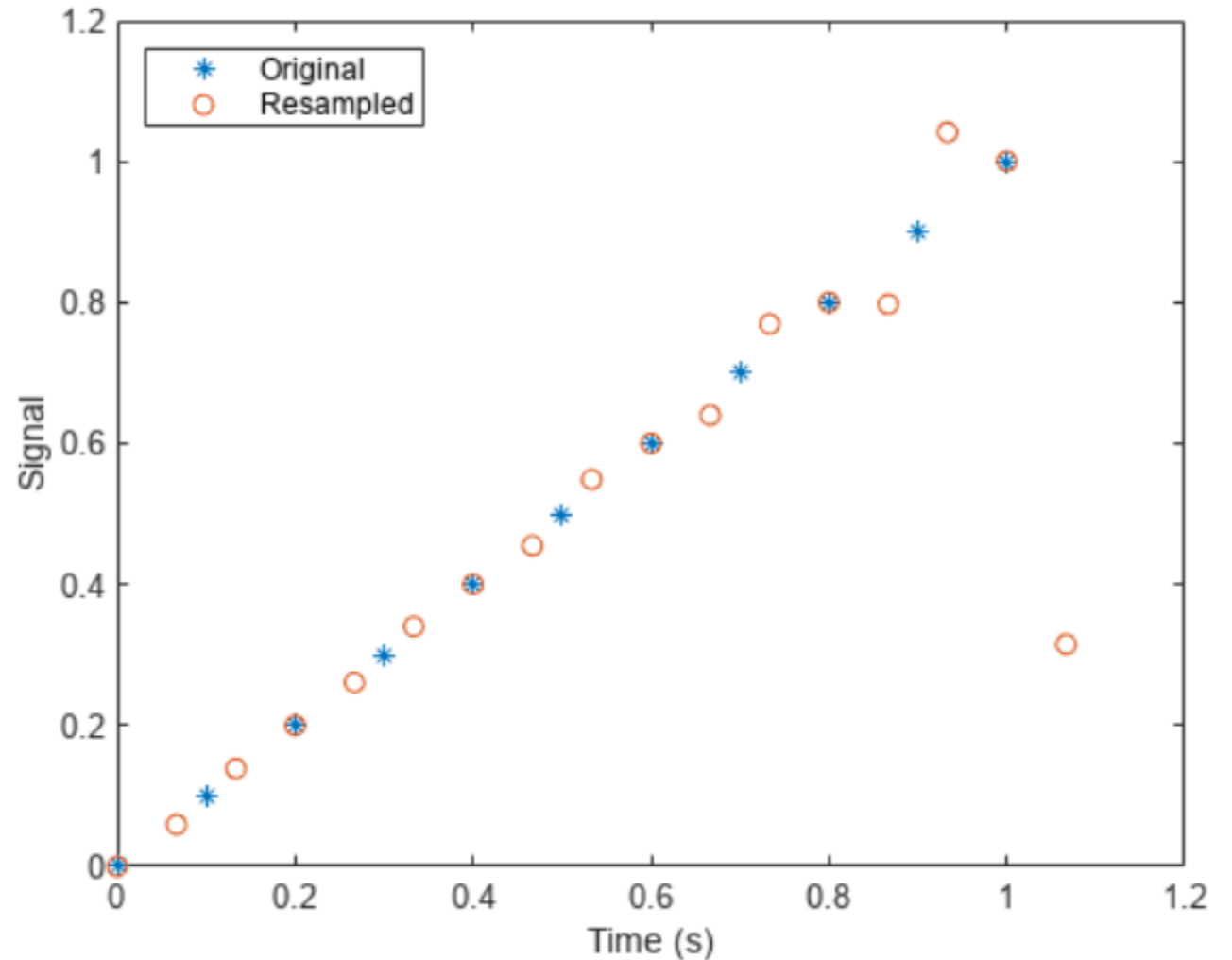
```
yTT = resample(xTT,p,q, __ )
```

```
y = resample(x,tx)
y = resample(x,tx,fs)
y = resample(x,tx,fs,p,q)
y = resample(x,tx, __ ,method)
[y,ty] = resample(x,tx, __ )
```

```
yTT = resample(xTT, __ )
```

```
[y,b] = resample(x,p,q, __ )
[y,ty,b] = resample(x,tx, __ )
[yTT,b] = resample(xTT, __ )
```

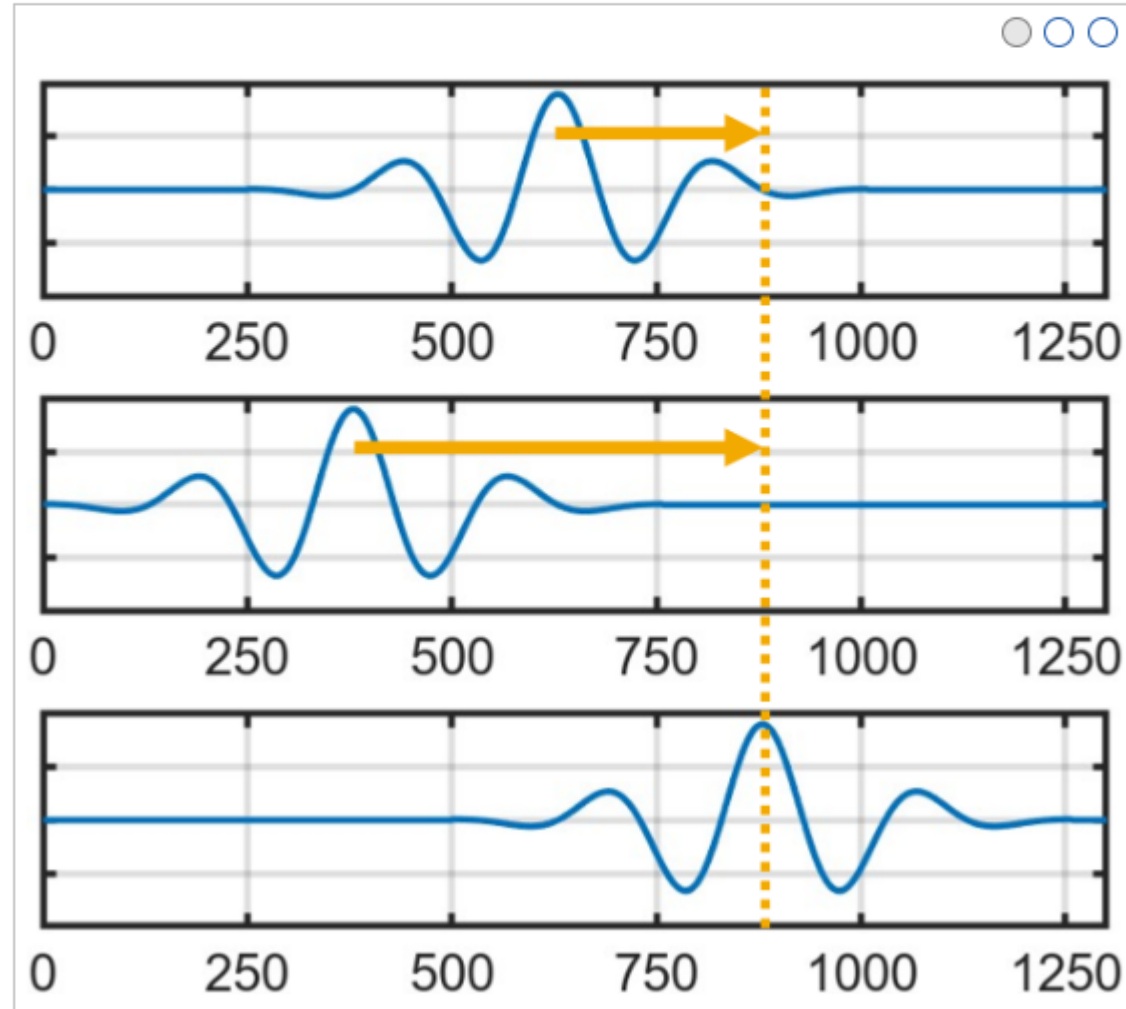
```
[ __ ] = resample( __ ,Dimension=dim)
```



Align Signals

The seismic stations are located in different places in Alaska, so the surface waves from the earthquake in Indonesia arrived at different times. This caused a lag between the recorded waves.

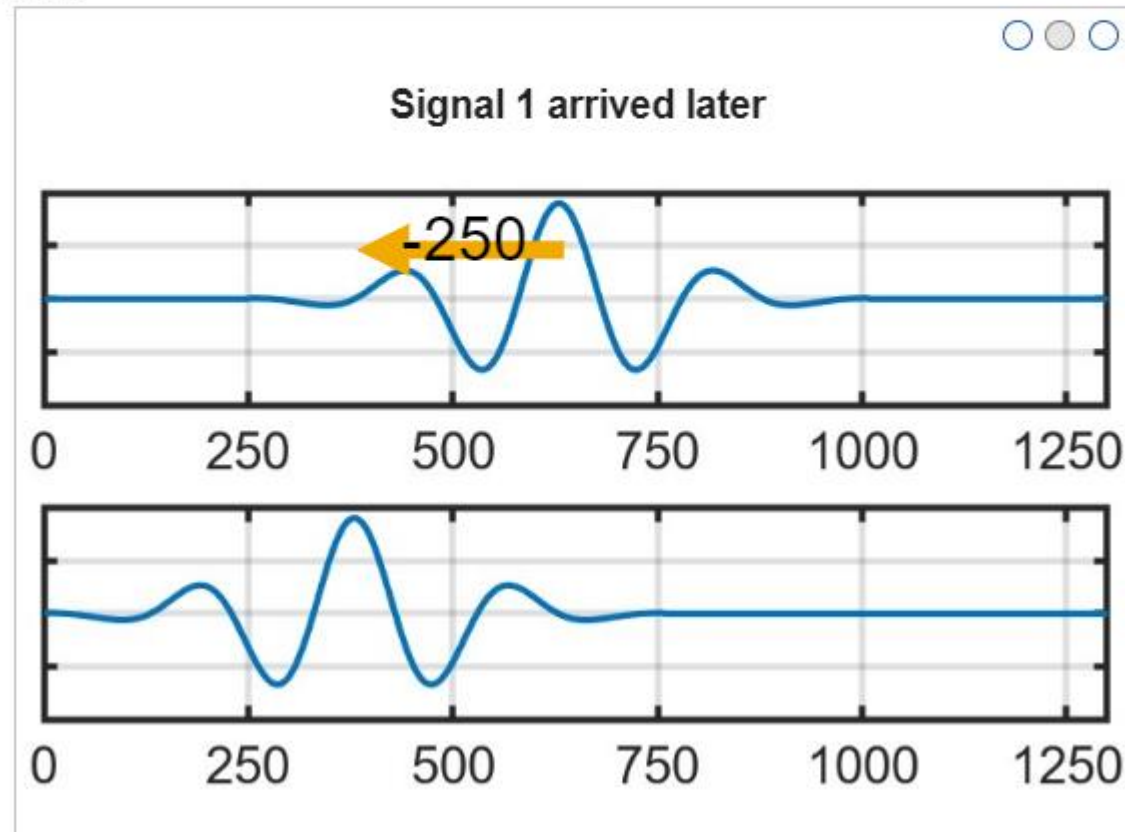
The `alignsignals` function aligns two signals by delaying the earlier signal. How do you align more than two signals?



Align Signals

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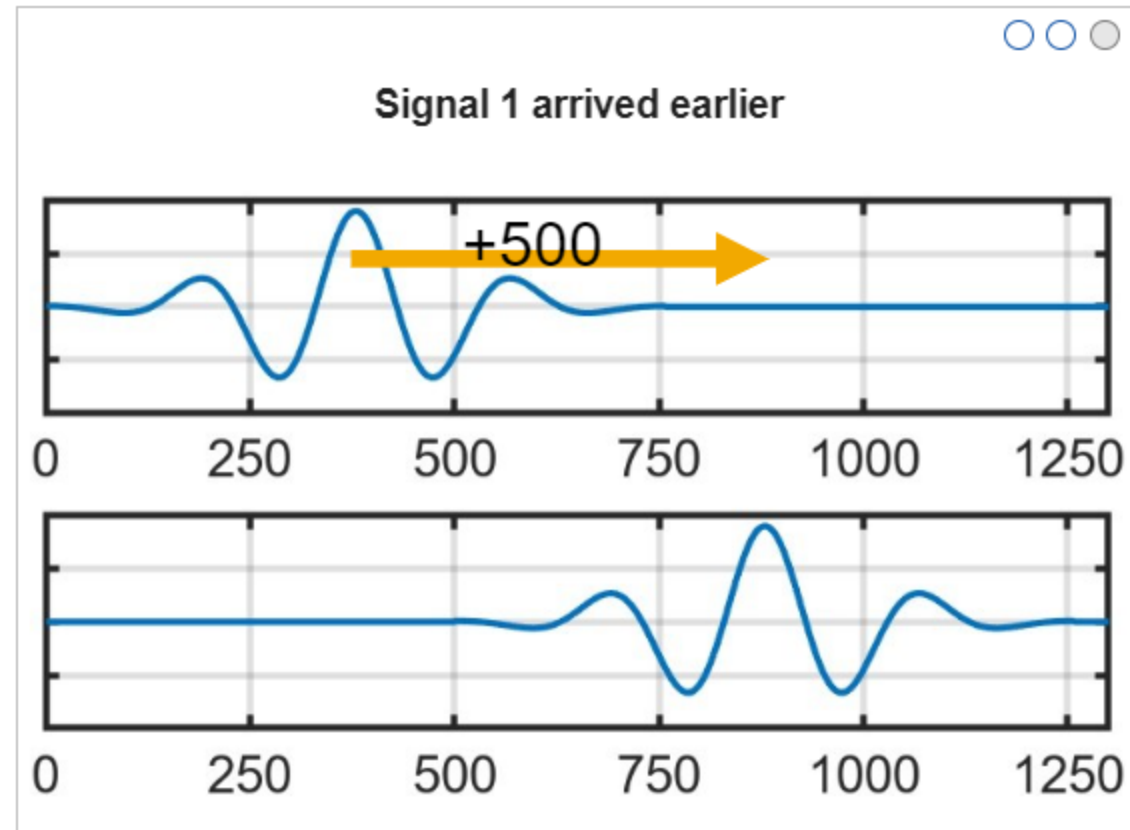
The `alignsignals` function aligns two signals by delaying the earlier signal. How do you align more than two signals?



Align Signals

The seismic stations are located in different places in Alaska, so the surface waves from the earthquake in Indonesia arrived at different times. This caused a lag between the recorded waves.

The `alignsignals` function aligns two signals by delaying the earlier signal. How do you align more than two signals?



Align Signals

Use the `finddelay` function to find the signal that arrived last.

```
finddelay(harp,pax)  
finddelay(harp,wanc)  
finddelay(pax,wanc)
```

Based on the delay values to the right, you can conclude that:

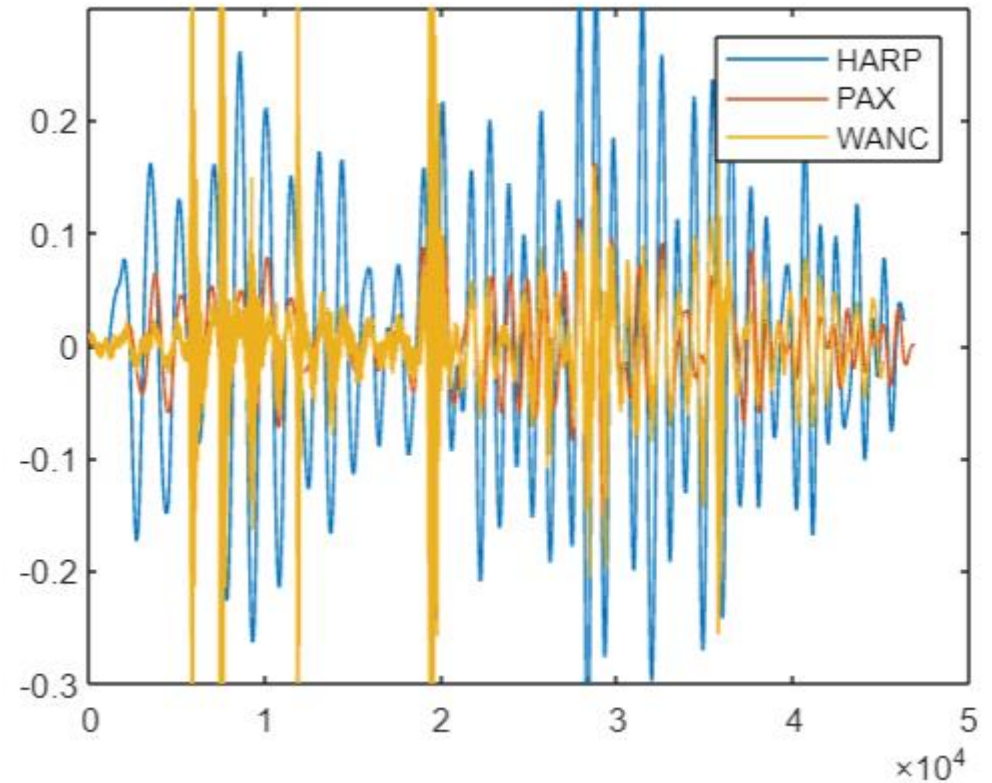
- harp arrived 596 samples after pax
- harp arrived 1167 samples before wanc
- pax arrived 1770 samples before wanc

So the wanc signal arrived last.

Since all three seismic stations should have recorded the same surface waves, all three signals should be correlated. The `alignsignals` function uses cross-correlation to estimate and delay signals.

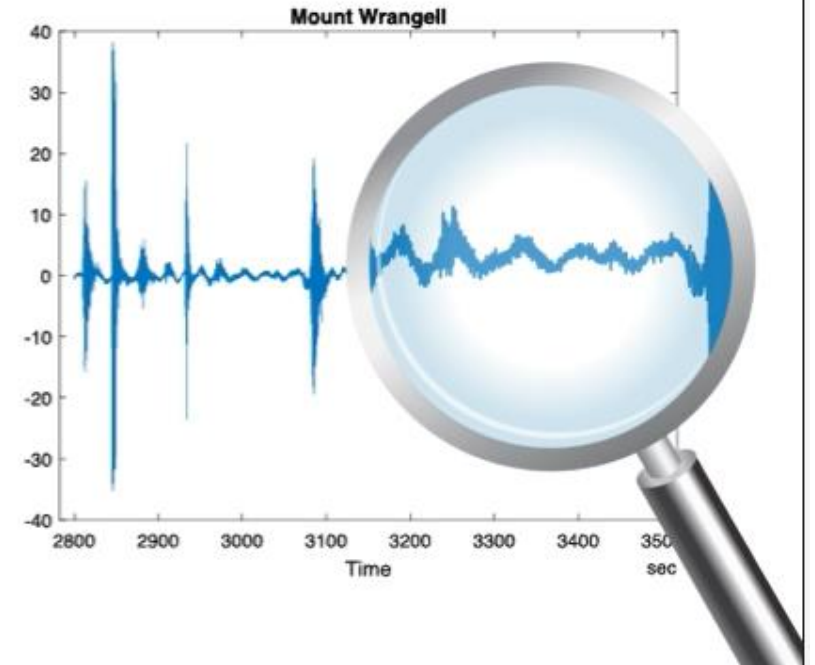
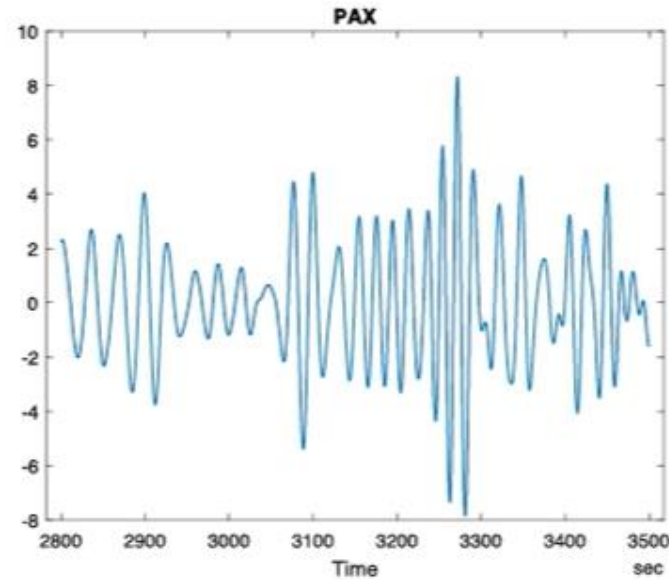
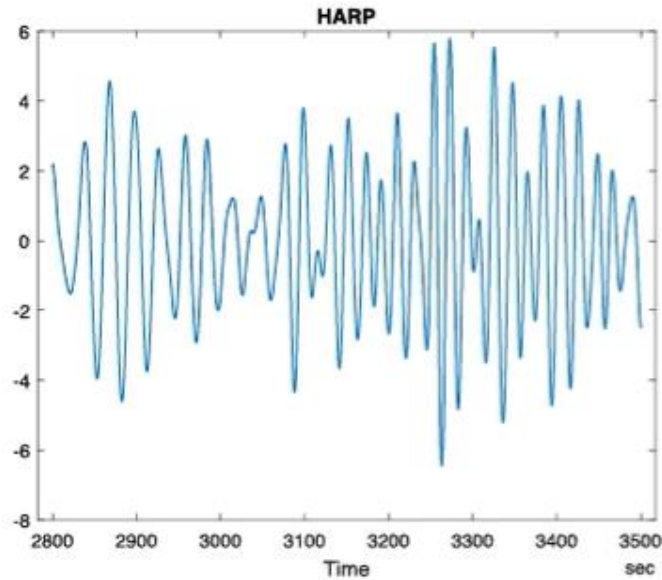
```
[xa,ya] = alignsignals(x,y)
```

If two signals have a sufficient correlation, the earlier output signal is padded with zeros.



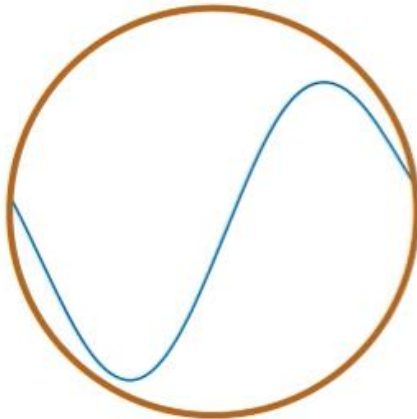
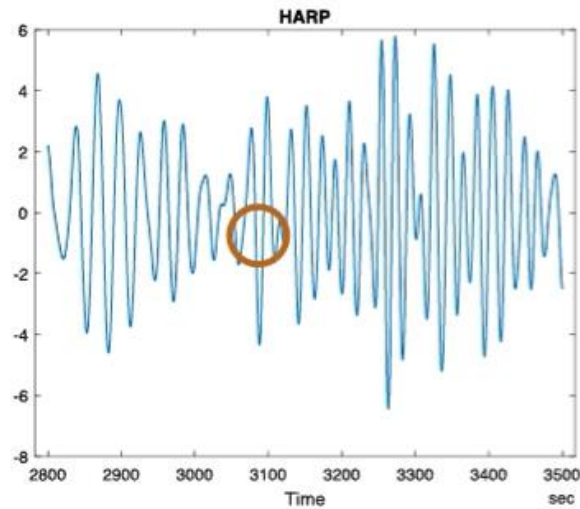
Viewing signals only in time domain is limited

Why these signals looks different at three locations?

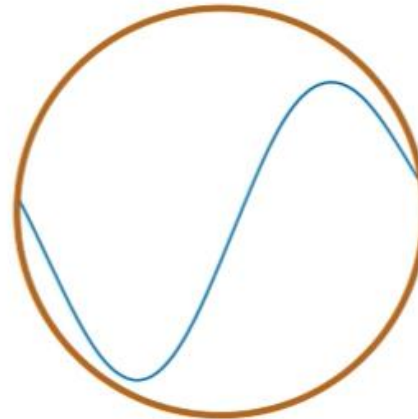
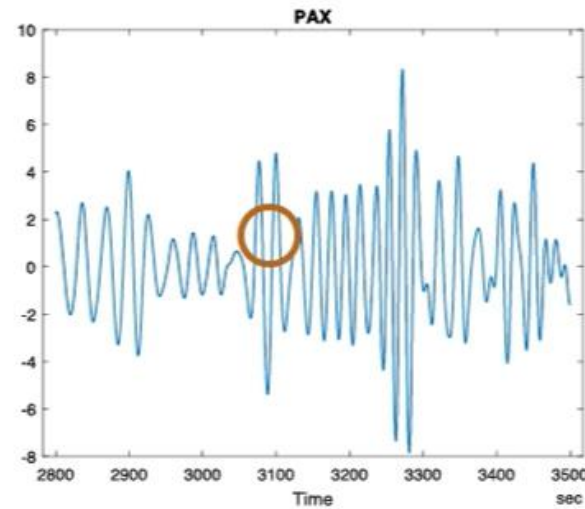


Viewing signals only in time domain is limited

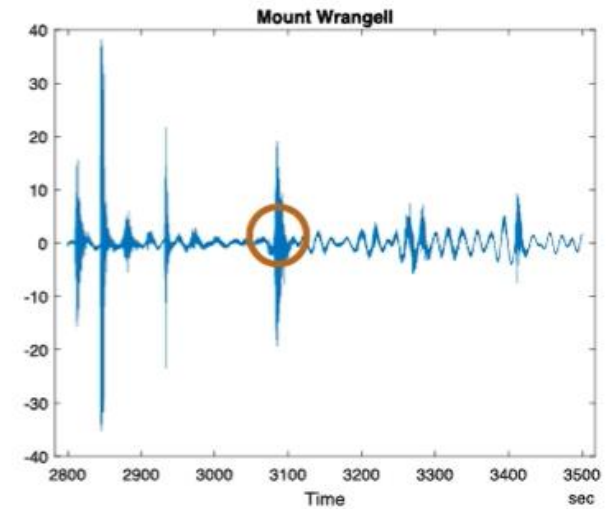
Low frequencies



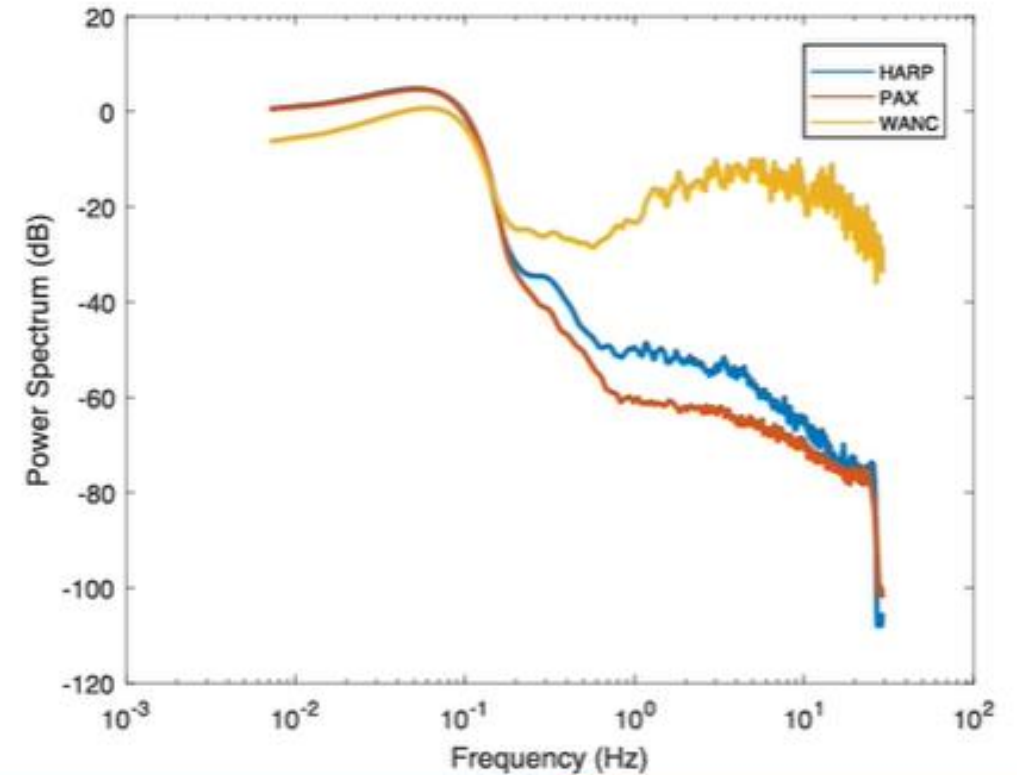
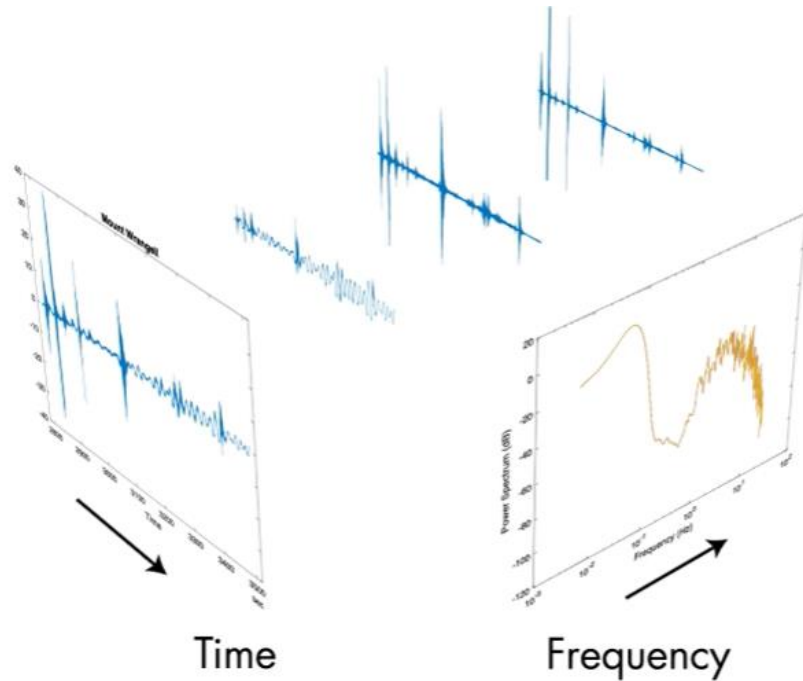
Low frequencies



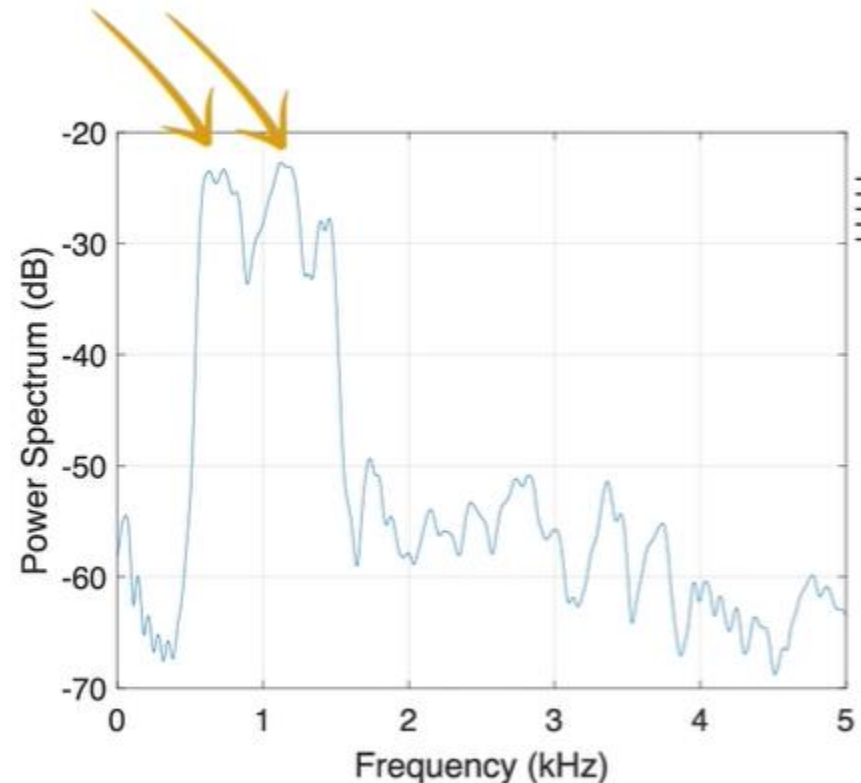
Low frequencies + high frequencies



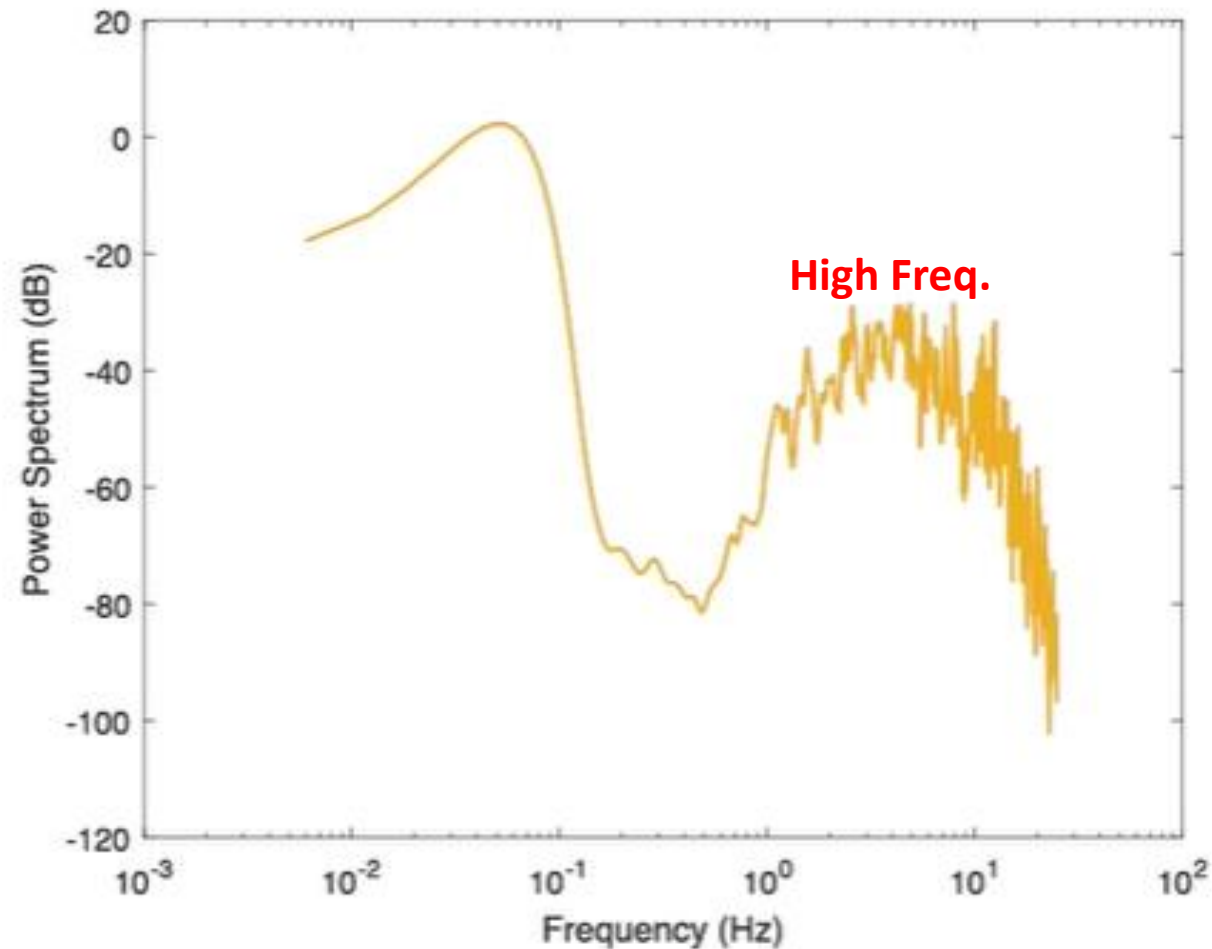
The standard way to decompose a signal into its frequency components is to use Fourier transforms



**Power spectrum shows the frequency content for the entire signal ,
but a valuable information could be missing about when different
frequencies occurred**

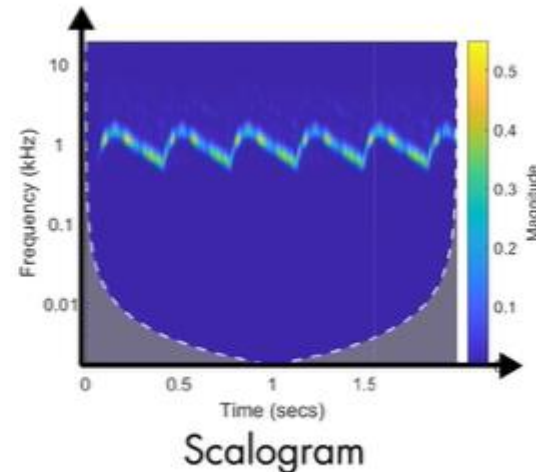
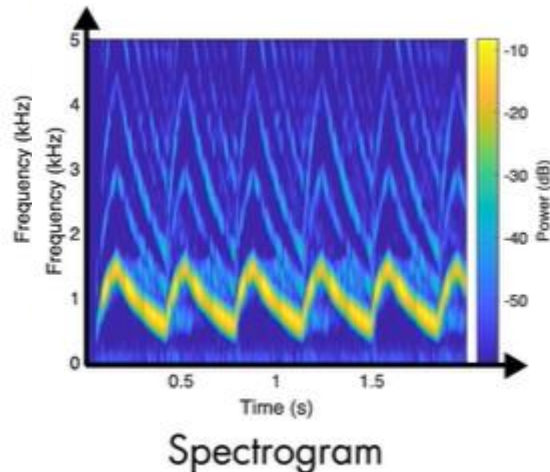
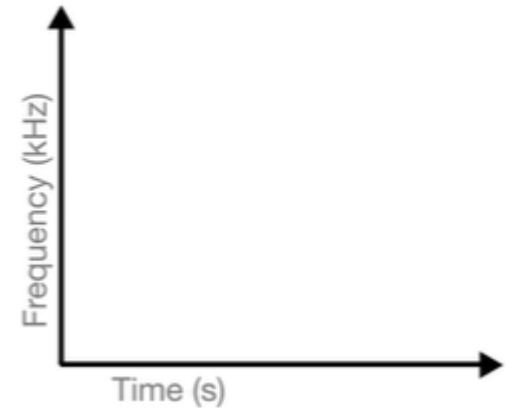


In the following signals , we cannot tell when high freq. occurs



Thus, we need a Time-Frequency Analysis

- Tells us information for both time and frequency domains
- Spectrograms and scalograms are two standard ways to do time frequency analysis
- Both decompose a signal into a combination of simple functions

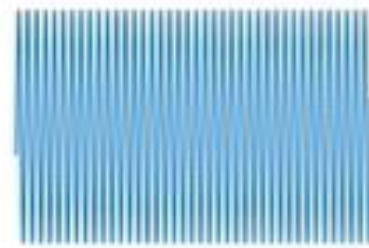
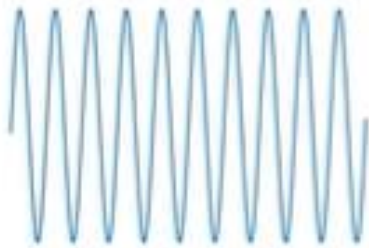
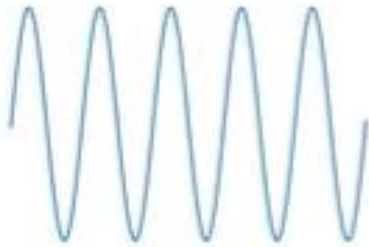


Spectrograms and scalograms

Fourier (sines)

Wavelets

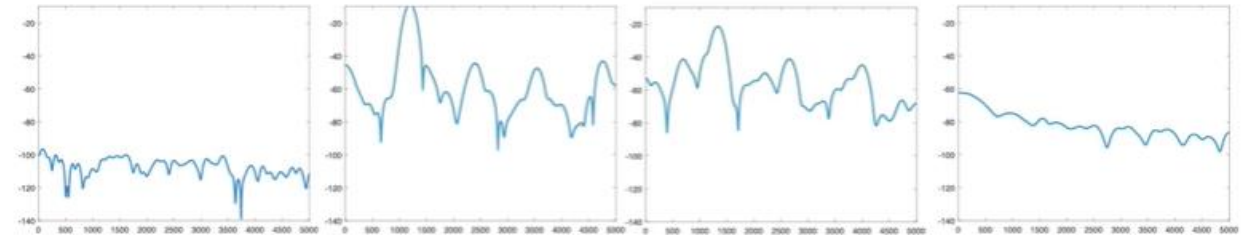
Spectrogram



scalograms

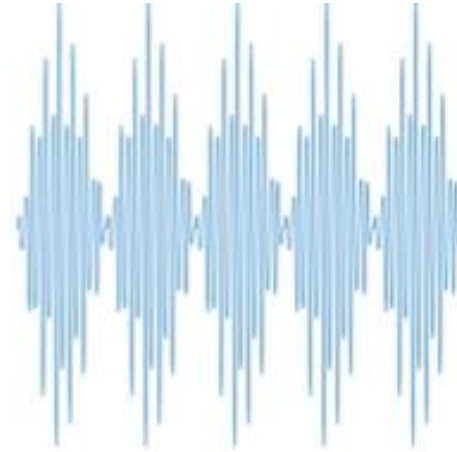
Spectrograms and scalograms: Principle of working

- The signal is sliced into short blocks
- Then, the spectrum is calculated for each block



Spectrograms and scalogram: How to know which one to use?

Spectrogram

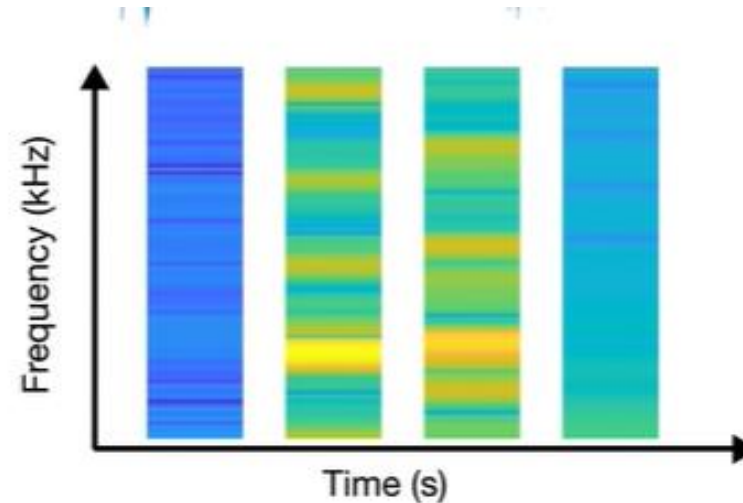
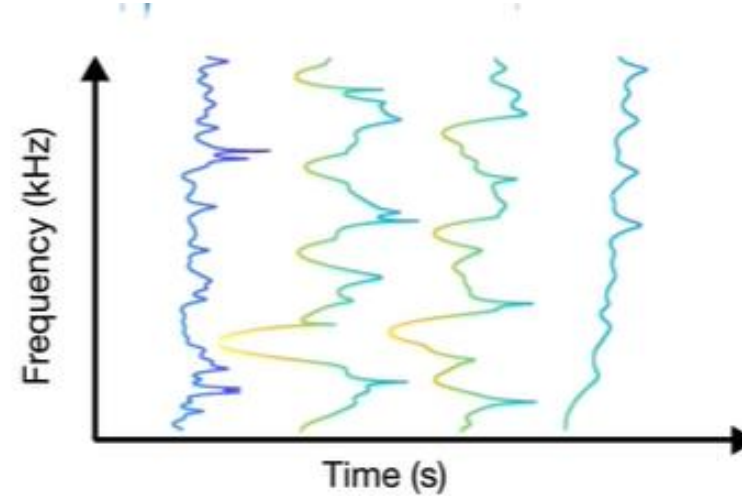


Scalogram



Spectrograms and scalograms: Principle of working

- The power of each frequency is represented by color
- Each pixel represents the amount of that frequency present at that time



References

- [1] Apolinário, Isabela F., and Paulo SR Diniz. "Introduction to signal processing theory." In *Academic press library in signal processing*, vol. 1, pp. 3-28. Elsevier, 2014.